UNDERGRADUATE PROJECT

Structure-Preserving Numerical Methods for Ordinary Differential Equations

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Abstract

This project studies structure-preserving numerical methods for Hamiltonian systems. It examines how relaxation and symplecticity affect the conservation of energy and phase-space area. The project first presents the Hamiltonian formulation of classical mechanics and the mathematical basis of symplectic methods. It then compares standard Runge–Kutta methods, symplectic methods, and relaxation-based methods through numerical experiments on the mathematical pendulum. The experiments measure changes in the Hamiltonian and in the area of a region in phase space.

The results show that relaxation improves phase-space area conservation, while symplecticity improves Hamiltonian conservation to some extent. Relaxation methods conserve energy to the precision of the nonlinear solver. Symplectic methods preserve phase-space area in theory, with small errors in its numerical computation. These findings suggest that combining relaxation and symplecticity may offer a way to improve the conservation of both quantities. However, this possibility requires further investigation to determine its effectiveness and limitations.

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1 From Lagrangian to Hamiltonian Mechanics

The position of a mechanical system with d degrees of freedom is typically described by a vector of generalized coordinates $q = (q_1, \dots, q_d)^T$. The dynamics of such a system are built upon two fundamental quantities: the kinetic energy $T = T(q, \dot{q})$ and the potential energy $U = U(q)$. By taking the difference of these energies, we construct the Lagrangian of the system:

$$L(q, \dot{q}) = T(q, \dot{q}) - U(q) \tag{1}$$

The motion of the system is then determined by the Euler-Lagrange equations:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) = \frac{\partial L}{\partial q} \tag{2}$$

While these equations successfully model a vast array of physical systems, they inherently constitute a system of second-order differential equations.

To simplify the structure and transform this framework into a highly symmetric system of first-order differential equations, we introduce a new set of independent variables called Poisson's variables, or the conjugate momenta, defined as:

$$p_k = \frac{\partial L}{\partial \dot{q}_k}(q, \dot{q}) \quad \text{for } k = 1, \dots, d \tag{3}$$

This transition mathematically relies on the Legendre transform. Specifically, it is required that the definition of the momenta establishes a continuously differentiable bijection $\dot{q} \leftrightarrow p$ between the velocities $\dot{q}$ and the momenta p for every configuration q. This critical requirement ensures that the relationship is invertible, allowing us to explicitly express the velocities as a function of the coordinates and momenta, $\dot{q} = \dot{q}(p, q)$.

With this inversion established, we can rigorously define the Hamiltonian function H as a function of p and q by substituting $\dot{q}(p, q)$ into the transformed energy equation:

$$H(p, q) = p^T \dot{q}(p, q) - L(q, \dot{q}(p, q)) \tag{4}$$

Theorem 1.1 (Equivalence of Lagrange and Hamilton Equations)**.** *Lagrange's equations are equivalent to Hamilton's canonical equations:*

$$\dot{p}_k = -\frac{\partial H}{\partial q_k}, \quad \dot{q}_k = \frac{\partial H}{\partial p_k} \tag{5}$$

Proof. By definition, the Hamiltonian is a function of p and q, meaning we must treat $\dot{q}$ as a function of p and q. When we take the partial derivatives of H with respect to p and q using the product rule and chain rule, we obtain:

$$\frac{\partial H}{\partial p} = \dot{q}^T + p^T \frac{\partial \dot{q}}{\partial p} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p} = \dot{q}^T \tag{6}$$

$$\frac{\partial H}{\partial q} = p^T \frac{\partial \dot{q}}{\partial q} - \frac{\partial L}{\partial q} - \frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial q} = -\frac{\partial L}{\partial q} \tag{7}$$

Notice how the terms $p^T \frac{\partial \dot{q}}{\partial p}$ and $\frac{\partial L}{\partial \dot{q}} \frac{\partial \dot{q}}{\partial p}$ perfectly cancel each other out precisely because we defined our momenta as $p = \frac{\partial L}{\partial \dot{q}}$. The exact same cancellation occurs in the second equation.

To fully understand why this establishes equivalence, we must look at the theoretical implications of these two resulting derivatives.

The first result, $\frac{\partial H}{\partial p} = \dot{q}^T$ (or simply $\dot{q} = \frac{\partial H}{\partial p}$), is a kinematic relation. It inherently satisfies the definition of our conjugate momenta and ensures that our variable substitution via the Legendre transform remains consistent.

The second result yields $\frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q}$. If we substitute the original second-order Euler-Lagrange equation, $\frac{\partial L}{\partial q} = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right)$, and recall our momenta definition $p = \frac{\partial L}{\partial \dot{q}}$, the right side becomes precisely $-\dot{p}$. Therefore, we arrive at $\dot{p} = -\frac{\partial H}{\partial q}$.

Together, these steps demonstrate that the original second-order Lagrangian system is decoupled into the symmetric pair of first-order Hamiltonian equations. □

1.1 Case of Quadratic Kinetic Energy (T)

In many classical mechanical systems, the kinetic energy is a quadratic function of the velocities. Specifically, suppose that $T = \frac{1}{2}\dot{q}^T M(q)\dot{q}$, where the mass matrix $M(q)$ is a symmetric and positive definite matrix.

Recalling that the Lagrangian is $L(q, \dot{q}) = T(q, \dot{q}) - U(q)$, and noting that the potential energy $U(q)$ is independent of velocity, the conjugate momentum is computed as the gradient of the kinetic energy with respect to $\dot{q}$. Because $M(q)$ is symmetric, this yields:

$$p = \nabla_{\dot{q}} L = M(q)\dot{q} \tag{8}$$

For a fixed configuration q, the positive definiteness of $M(q)$ ensures that this matrix is non-singular and invertible. Consequently, this relationship strictly satisfies the conditions for the existence of the Legendre transform. We can explicitly invert this relation to solve for the velocities in terms of the momenta:

$$\dot{q} = M(q)^{-1} p \tag{9}$$

To construct the Hamiltonian $H(p, q)$, we substitute this inverted velocity relation into the fundamental definition $H(p, q) = p^T \dot{q} - L(q, \dot{q})$:

$$H(p, q) = p^T M(q)^{-1} p - L(q, M(q)^{-1} p) \tag{10}$$

Expanding the Lagrangian term $L = T - U$ using our momentum substitution:

$$H(p, q) = p^T M(q)^{-1} p - \left(\frac{1}{2} (M(q)^{-1} p)^T M(q) (M(q)^{-1} p) - U(q) \right) \tag{11}$$

$$= p^T M(q)^{-1} p - \frac{1}{2} p^T (M(q)^{-1})^T M(q) M(q)^{-1} p + U(q) \tag{12}$$

Because $M(q)$ is symmetric, its inverse $M(q)^{-1}$ is also symmetric, meaning $(M(q)^{-1})^T = M(q)^{-1}$. Since $M(q)M(q)^{-1} = I$, the kinetic energy term simplifies cleanly:

$$H(p, q) = p^T M(q)^{-1} p - \frac{1}{2} p^T M(q)^{-1} p + U(q) \tag{13}$$

$$= \frac{1}{2} p^T M(q)^{-1} p + U(q) \tag{14}$$

Recognizing that the first term is exactly the kinetic energy T expressed in terms of momentum, we arrive at the fundamental physical interpretation:

$$H = T + U \tag{15}$$

Therefore, for systems with quadratic kinetic energy, the Hamiltonian is identically the total energy of the mechanical system.

Example 1.2 (The Mathematical Pendulum)**.** To illustrate this transition physically and mathematically, consider the simple mathematical pendulum. We define its position using a single generalized coordinate, the angle $q = \alpha$.

The kinetic energy T and potential energy U of the pendulum are given by:

$$T = \frac{1}{2} m l^2 \dot{q}^2, \quad U = -mgl \cos q \tag{16}$$

where m is the mass, l is the length of the pendulum, and g is the acceleration due to gravity. The Lagrangian $L = T - U$ is therefore:

$$L(q, \dot{q}) = \frac{1}{2} m l^2 \dot{q}^2 + mgl \cos q \tag{17}$$

Applying the Euler-Lagrange equation, $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}}\right) = \frac{\partial L}{\partial q}$, we obtain the second-order differential equation of motion:

$$ml^2 \ddot{q} = -mgl \sin q \implies \ddot{q} + \frac{g}{l} \sin q = 0 \tag{18}$$

This is the classic formulation of the pendulum's dynamics. To transform this into a Hamiltonian system, we first simplify the expressions by applying the normalization $m = l = g = 1$. Our normalized Lagrangian is $L = \frac{1}{2}\dot{q}^2 + \cos q$.

We introduce the conjugate momentum p, which represents the angular momentum:

$$p = \frac{\partial L}{\partial \dot{q}} = \dot{q} \tag{19}$$

Using the Legendre transform to eliminate $\dot{q}$, we construct the Hamiltonian function $H(p, q)$:

$$H(p, q) = p\dot{q} - L = p(p) - \left(\frac{1}{2}p^2 + \cos q\right) = \frac{1}{2}p^2 - \cos q \tag{20}$$

Applying Hamilton's canonical equations, we recover the decoupled first-order system:

$$\dot{q} = \frac{\partial H}{\partial p} = p, \quad \dot{p} = -\frac{\partial H}{\partial q} = -\sin q \tag{21}$$

Physically, because the original kinetic energy T was quadratic with respect to the velocities, the resulting Hamiltonian $H = T + U$ represents the total mechanical energy of the system. The Hamiltonian formulation thus shifts our perspective: rather than viewing the pendulum's motion merely as a second-order response to restoring forces (the Lagrangian view), we now understand its evolution as a geometric flow through phase space (p, q) that strictly conserves the total energy, $H(p, q)$.

2 Symplectic Transformations

2.1 Oriented Area

To understand the geometric properties of Hamiltonian systems, we first consider a two-dimensional parallelogram P in phase space spanned by two vectors $\xi = (\xi^p, \xi^q)^T$ and $\eta = (\eta^p, \eta^q)^T$.

In one dimension ($d = 1$), the oriented area of this parallelogram is simply the determinant of the coordinates:

$$\text{or.area}(P) = \det \begin{pmatrix} \xi^p & \eta^p \\ \xi^q & \eta^q \end{pmatrix} = \xi^p \eta^q - \xi^q \eta^p \tag{22}$$

For higher dimensions ($d > 1$), we generalize this by taking the sum of the oriented areas of the projections of the parallelogram onto the coordinate planes (p_i, q_i). This is expressed using the bilinear mapping $\omega(\xi, \eta)$:

$$\omega(\xi, \eta) := \sum_{i=1}^{d} \det \begin{pmatrix} \xi_i^p & \eta_i^p \\ \xi_i^q & \eta_i^q \end{pmatrix} = \sum_{i=1}^{d} (\xi_i^p \eta_i^q - \xi_i^q \eta_i^p) \tag{23}$$

In matrix notation, this takes the elegant form $\omega(\xi, \eta) = \xi^T J \eta$, where $J = \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix}$ and I is the $d \times d$ identity matrix.

2.2 Defining Symplectic Maps

Definition 2.1 (Symplectic Linear Mapping)**.** A linear mapping $A : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ is called *symplectic* if

$$A^T J A = J \tag{24}$$

or, equivalently, if $\omega(A\xi, A\eta) = \omega(\xi, \eta)$ for all $\xi, \eta \in \mathbb{R}^{2d}$.

Example 2.2 (2D Phase Space and Area Preservation)**.** To illustrate this definition in the simplest case, consider a 2D phase space ($d = 1$) with coordinates (p, q). A linear transformation A on this space is symplectic if and only if $\det(A) = 1$.

Proof. Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ be a general 2×2 transformation matrix, and recall the standard symplectic structure matrix for $d = 1$ is $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. We evaluate the left-hand side of the symplecticity condition, $A^T J A$, step-by-step. First, computing the product JA:

$$JA = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} \tag{25}$$

Next, multiplying this result by A^T from the left yields:

$$A^T J A = \begin{pmatrix} a & c \\ b & d \end{pmatrix} \begin{pmatrix} c & d \\ -a & -b \end{pmatrix} = \begin{pmatrix} ac - ac & ad - bc \\ bc - ad & bd - bd \end{pmatrix} \tag{26}$$

Simplifying the diagonal terms, we obtain:

$$A^T J A = \begin{pmatrix} 0 & ad - bc \\ -(ad - bc) & 0 \end{pmatrix} \tag{27}$$

Recognizing that the determinant of A is precisely $\det(A) = ad - bc$, we can factor this scalar value out of the matrix:

$$A^T J A = \det(A) \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \det(A) J \tag{28}$$

We now substitute this universal 2×2 identity directly into our defining symplecticity condition:

$$A^T J A = J \iff \det(A) J = J \tag{29}$$

Since the structure matrix J is non-zero, we can equate the scalar coefficients. This equation holds true if and only if $\det(A) = 1$. □

This concept extends to non-linear maps by examining their local linear approximations (their Jacobians).

Definition 2.3 (Symplectic Differentiable Mapping)**.** A differentiable map $g : U \to \mathbb{R}^{2d}$ (where $U \subset \mathbb{R}^{2d}$ is an open set) is called *symplectic* if the Jacobian matrix $g'(p, q)$ is everywhere symplectic, i.e., if

$$g'(p, q)^T J g'(p, q) = J \quad \text{or} \quad \omega(g'(p, q)\xi, g'(p, q)\eta) = \omega(\xi, \eta) \tag{30}$$

2.3 Geometric Interpretation on Manifolds

We can geometrically interpret this by considering a 2-dimensional sub-manifold M defined as the image $M = \psi(K)$ of a compact set $K \subset \mathbb{R}^2$, where $\psi(s, t)$ is a continuously differentiable parameterization.

If we divide the compact set K into infinitesimal compartments $ds\,dt$, the manifold M can be approximated as the union of small parallelograms spanned by the tangent vectors $\frac{\partial \psi}{\partial s}(s, t)\,ds$ and $\frac{\partial \psi}{\partial t}(s, t)\,dt$. Summing the oriented areas of the projections of all these infinitesimal parallelograms yields the total oriented area over the manifold:

$$\Omega(M) = \iint_K \omega \left(\frac{\partial \psi}{\partial s}(s, t), \frac{\partial \psi}{\partial t}(s, t) \right) ds\,dt \tag{31}$$

Lemma 2.4 (Area Preservation on 2-Manifolds)**.** *If the mapping $g : U \to \mathbb{R}^{2d}$ is symplectic on U, then it preserves the expression $\Omega(M)$, i.e.,*

$$\Omega(g(M)) = \Omega(M) \tag{32}$$

holds for all 2-dimensional manifolds M that can be represented as the image of a continuously differentiable function ψ.

Proof. The transformed manifold $g(M)$ can be parameterized by the composition $g \circ \psi$. Setting up the integral for the area of this new manifold, we have:

$$\Omega(g(M)) = \iint_K \omega \left(\frac{\partial (g \circ \psi)}{\partial s}(s, t), \frac{\partial (g \circ \psi)}{\partial t}(s, t) \right) ds\,dt \tag{33}$$

By the chain rule, the partial derivatives are $\frac{\partial}{\partial s}(g \circ \psi) = g'(\psi(s, t)) \frac{\partial \psi}{\partial s}$ (and identically for t). Because g is a symplectic transformation, its Jacobian g' preserves the bilinear map ω, meaning $\omega(g'\xi, g'\eta) = \omega(\xi, \eta)$. Therefore, the integrand simplifies entirely, yielding:

$$\Omega(g(M)) = \iint_K \omega \left(\frac{\partial \psi}{\partial s}(s, t), \frac{\partial \psi}{\partial t}(s, t) \right) ds\,dt = \Omega(M) \tag{34}$$

This confirms that any symplectic mapping (including non-linear ones) is fundamentally area-preserving. □

3 Theorems on Symplectic Flow

Theorem 3.1 (Poincaré, 1899)**.** *Let $H(p,q)$ be a twice continuously differentiable function on $U \subset \mathbb{R}^{2d}$. Then, for each fixed t, the flow φ_t is a symplectic transformation wherever it is defined.*

Proof. Let $y = (p,q)$. We write the Hamiltonian system in the form $\dot{y} = J^{-1}\nabla H(y)$. The flow is given by $\varphi_t(y_0) = y(t)$.

We first derive the variational equation by differentiating the Jacobian of the flow, $\Psi = \frac{\partial \varphi_t}{\partial y_0}$, with respect to t:

$$\frac{d}{dt}\left(\frac{\partial \varphi_t}{\partial y_0}\right) = \frac{\partial}{\partial y_0}\left(\frac{d\varphi_t}{dt}\right) = \frac{\partial}{\partial y_0}\left(J^{-1}\nabla H(\varphi_t(y_0))\right) \tag{35}$$

Applying the chain rule yields:

$$\frac{d}{dt}\left(\frac{\partial \varphi_t}{\partial y_0}\right) = J^{-1}\nabla^2 H(\varphi_t(y_0))\frac{\partial \varphi_t}{\partial y_0} \tag{36}$$

where $\nabla^2 H$ is the symmetric Hessian matrix of H. Thus, Ψ is a solution to the variational equation $\dot{\Psi} = J^{-1}\nabla^2 H(\varphi_t(y_0))\Psi$.

To show the flow is symplectic, we differentiate the condition $\Psi^T J \Psi$ with respect to time:

$$\frac{d}{dt}\left(\Psi^T J \Psi\right) = \dot{\Psi}^T J \Psi + \Psi^T J \dot{\Psi} \tag{37}$$

Substituting our derived variational equation gives:

$$\frac{d}{dt}\left(\Psi^T J \Psi\right) = \left(J^{-1}\nabla^2 H(\varphi_t(y_0))\Psi\right)^T J\Psi + \Psi^T J\left(J^{-1}\nabla^2 H(\varphi_t(y_0))\Psi\right) \tag{38}$$

$$= \Psi^T \nabla^2 H(\varphi_t(y_0)) J^{-T} J \Psi + \Psi^T \nabla^2 H(\varphi_t(y_0))\Psi = 0 \tag{39}$$

because $J^T = -J$ and $J^{-T}J = -I$.

Since $\frac{d}{dt}\left(\Psi^T J\Psi\right) = 0$, the expression $\Psi^T J \Psi$ is constant for all t. At $t = 0$, φ_0 is the identity map, meaning $\Psi(0) = I$, which gives $I^T J I = J$. Therefore, $\left(\frac{\partial \varphi_t}{\partial y_0}\right)^T J \left(\frac{\partial \varphi_t}{\partial y_0}\right) = J$ for all t, proving the flow is symplectic. □

Definition 3.2 (Locally Hamiltonian System)**.** A differential equation $\dot{y} = f(y)$ is *locally Hamiltonian* if, for every $y_0 \in U$, there exists a neighbourhood where $f(y) = J^{-1}\nabla H(y)$ for some function H.

Theorem 3.3 (Characterization of Locally Hamiltonian Systems)**.** *Let $f : U \to \mathbb{R}^{2d}$ be continuously differentiable. Then, $\dot{y} = f(y)$ is locally Hamiltonian if and only if its flow $\varphi_t(y)$ is symplectic for all $y \in U$ and for all sufficiently small t.*

Proof. The necessity follows directly from Theorem 3.1. To prove sufficiency, we assume the flow φ_t is symplectic and must prove the local existence of a function $H(y)$ such that $f(y) = J^{-1}\nabla H(y)$.

We start by differentiating the symplecticity condition, $\left(\frac{\partial \varphi_t}{\partial y_0}\right)^T J \left(\frac{\partial \varphi_t}{\partial y_0}\right) = J$, with respect to t, knowing that $\frac{\partial \varphi_t}{\partial y_0}$ solves the variational equation $\dot{\Psi} = f'(\varphi_t(y_0))\Psi$:

$$\frac{d}{dt}\left(\left(\frac{\partial \varphi_t}{\partial y_0}\right)^T J \left(\frac{\partial \varphi_t}{\partial y_0}\right)\right) = \left(\frac{\partial \varphi_t}{\partial y_0}\right)^T \left(f'(\varphi_t(y_0))^T J + J f'(\varphi_t(y_0))\right)\left(\frac{\partial \varphi_t}{\partial y_0}\right) = 0 \tag{40}$$

Putting $t = 0$, where $\varphi_0(y_0) = y_0$ and $\frac{\partial \varphi_0}{\partial y_0} = I$, it follows from $J = -J^T$ that $Jf'(y_0)$ is a symmetric matrix for all y_0.

To find H, let us assume $y_0 = 0$ and consider a ball around 0 contained in U. On this ball, we construct the function $H(y)$ as:

$$H(y) = \int_0^1 y^T J f(ty)\, dt + \text{Const} \tag{41}$$

Let $g(y) = Jf(y)$; since its Jacobian $g'(y) = Jf'(y)$ is symmetric, we have $\frac{\partial g_i}{\partial y_k} = \frac{\partial g_k}{\partial y_i}$. Differentiating $H(y)$ with respect to y_k yields:

$$\frac{\partial H}{\partial y_k}(y) = \int_0^1 \left(g_k(ty) + y^T \frac{\partial g}{\partial y_k}(ty)t\right) dt \tag{42}$$

Using the symmetry property, the integrand simplifies as an exact derivative:

$$\frac{\partial H}{\partial y_k}(y) = \int_0^1 \frac{d}{dt}\left(t g_k(ty)\right) dt = g_k(y) \tag{43}$$

This proves that $\nabla H(y) = g(y) = Jf(y)$. Rearranging gives $f(y) = J^{-1}\nabla H(y)$, concluding the proof that the system is locally Hamiltonian. □

4 Symplecticity of the Symplectic Euler Method

Definition 4.1 (Symplectic Numerical Method)**.** A numerical one-step method is called *symplectic* if the one-step map $y_1 = \Phi_h(y_0)$ is symplectic whenever the method is applied to a smooth Hamiltonian system.

Theorem 4.2 (Symplecticity of the Symplectic Euler Method)**.** *The Symplectic Euler method, defined by the step equations:*

$$p_{n+1} = p_n - hH_q(p_{n+1}, q_n) \tag{44}$$
$$q_{n+1} = q_n + hH_p(p_{n+1}, q_n) \tag{45}$$

is a symplectic numerical method.

Proof. To prove that the Symplectic Euler method is indeed symplectic, we examine the Jacobian of the transformation.

Step 1: Implicit Differentiation

We first differentiate both equations with respect to the initial states p_n and q_n.

Block A: Differentiating $p_{n+1} = p_n - hH_q(p_{n+1}, q_n)$ 1. With respect to p_n:

$$\frac{\partial p_{n+1}}{\partial p_n} = I - hH_{qp}^T \frac{\partial p_{n+1}}{\partial p_n} - hH_{qq}\frac{\partial q_n}{\partial p_n}$$

Since $\frac{\partial q_n}{\partial p_n} = 0$, we get:

$$(I + hH_{qp}^T)\frac{\partial p_{n+1}}{\partial p_n} = I \tag{46}$$

2. With respect to q_n:

$$\frac{\partial p_{n+1}}{\partial q_n} = 0 - hH_{qp}^T \frac{\partial p_{n+1}}{\partial q_n} - hH_{qq}\frac{\partial q_n}{\partial q_n}$$

Since $\frac{\partial q_n}{\partial q_n} = I$, we get:

$$(I + hH_{qp}^T)\frac{\partial p_{n+1}}{\partial q_n} = -hH_{qq} \tag{47}$$

Block B: Differentiating $q_{n+1} = q_n + hH_p(p_{n+1}, q_n)$ 1. With respect to p_n:

$$\frac{\partial q_{n+1}}{\partial p_n} = \frac{\partial q_n}{\partial p_n} + hH_{pp}\frac{\partial p_{n+1}}{\partial p_n} + hH_{qp}\frac{\partial q_n}{\partial p_n}$$

Since $\frac{\partial q_n}{\partial p_n} = 0$, we get:

$$-hH_{pp}\frac{\partial p_{n+1}}{\partial p_n} + \frac{\partial q_{n+1}}{\partial p_n} = 0 \tag{48}$$

2. With respect to q_n:

$$\frac{\partial q_{n+1}}{\partial q_n} = I + hH_{pp}\frac{\partial p_{n+1}}{\partial q_n} + hH_{qp}\frac{\partial q_n}{\partial q_n}$$

Since $\frac{\partial q_n}{\partial q_n} = I$, we get:

$$-hH_{pp}\frac{\partial p_{n+1}}{\partial q_n} + \frac{\partial q_{n+1}}{\partial q_n} = I + hH_{qp} \tag{49}$$

Step 2: The Compact Block Matrix Formulation

Combining Equations (46), (47), (48), and (49) into a single block matrix system yields:

$$\begin{pmatrix} I + hH_{qp}^T & 0 \\ -hH_{pp} & I \end{pmatrix} \begin{pmatrix} \frac{\partial p_{n+1}}{\partial p_n} & \frac{\partial p_{n+1}}{\partial q_n} \\ \frac{\partial q_{n+1}}{\partial p_n} & \frac{\partial q_{n+1}}{\partial q_n} \end{pmatrix} = \begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{qp} \end{pmatrix} \tag{50}$$

Let us define the helper matrices A and B:

$$A = \begin{pmatrix} I + hH_{qp}^T & 0 \\ -hH_{pp} & I \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} I & -hH_{qq} \\ 0 & I + hH_{qp} \end{pmatrix} \tag{51}$$

Thus, we can write the relation compactly as:

$$AG = B \implies G = A^{-1}B \tag{52}$$

Step 3: Invertibility and Transpose Symplecticity of the Step Map

For a sufficiently small step size h (specifically whenever $\|hH_{qp}\| < 1$), the block matrices A and B are invertible. Consequently, the step Jacobian matrix $G = A^{-1}B$ is strictly invertible.

Because G is invertible in our scenario, the standard symplecticity condition $G^T J G = J$ can be equivalently evaluated in its transpose form $GJG^T = J$:

1. Since G is invertible, take the matrix inverse of both sides of $G^T J G = J$:
$$(G^T J G)^{-1} = J^{-1}$$
2. Expand the left-hand side inverse using the product rule $(ABC)^{-1} = C^{-1}B^{-1}A^{-1}$:
$$G^{-1}J^{-1}(G^T)^{-1} = J^{-1}$$
3. Substitute $J^{-1} = -J$:
$$G^{-1}(-J)(G^T)^{-1} = -J \implies G^{-1}J(G^T)^{-1} = J$$
4. Multiply on the left by G and on the right by G^T:
$$G\left[G^{-1}J(G^T)^{-1}\right]G^T = GJG^T \implies J = GJG^T$$

Thus, for our invertible step map G, establishing $GJG^T = J$ is equivalent to showing $G^T J G = J$.

Substituting $G = A^{-1}B$ and $G^T = B^T A^{-T}$ into $GJG^T = J$ yields:

$$(A^{-1}B)J(B^T A^{-T}) = J \implies A^{-1}(BJB^T)A^{-T} = J$$

Since A and A^T are invertible, multiplying on the left by A and on the right by A^T gives:

$$A\left[A^{-1}(BJB^T)A^{-T}\right]A^T = AJA^T \implies BJB^T = AJA^T \implies AJA^T = BJB^T$$

This proves that establishing $AJA^T = BJB^T$ is strictly equivalent to showing that the Symplectic Euler method is symplectic.

Part A: Calculating AJA^T Let $X = I + hH_{qp}^T$ and $Y = -hH_{pp}$ (where $Y^T = Y$ due to symmetry):

$$A = \begin{pmatrix} X & 0 \\ Y & I \end{pmatrix}$$

$$AJ = \begin{pmatrix} X & 0 \\ Y & I \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \begin{pmatrix} 0 & X \\ -I & Y \end{pmatrix}$$

$$AJA^T = \begin{pmatrix} 0 & X \\ -I & Y \end{pmatrix} \begin{pmatrix} X^T & Y^T \\ 0 & I \end{pmatrix} = \begin{pmatrix} 0 & X \\ -X^T & -Y^T + Y \end{pmatrix}$$

Since Y is symmetric, the bottom-right term $-Y^T + Y = 0$. Substituting X back in yields:

$$AJA^T = \begin{pmatrix} 0 & I + hH_{qp}^T \\ -(I + hH_{qp}) & 0 \end{pmatrix} \tag{53}$$

Part B: Calculating BJB^T Let $Z = -hH_{qq}$ (where $Z^T = Z$ due to symmetry) and $W = I + hH_{qp}$:

$$B = \begin{pmatrix} I & Z \\ 0 & W \end{pmatrix}$$

$$BJ = \begin{pmatrix} I & Z \\ 0 & W \end{pmatrix} \begin{pmatrix} 0 & I \\ -I & 0 \end{pmatrix} = \begin{pmatrix} -Z & I \\ -W & 0 \end{pmatrix}$$

$$BJB^T = \begin{pmatrix} -Z & I \\ -W & 0 \end{pmatrix} \begin{pmatrix} I & 0 \\ Z^T & W^T \end{pmatrix} = \begin{pmatrix} -Z + Z^T & W^T \\ -W & 0 \end{pmatrix}$$

Since Z is symmetric, the top-left term $-Z + Z^T = 0$. Substituting W back in yields:

$$BJB^T = \begin{pmatrix} 0 & I + hH_{qp}^T \\ -(I + hH_{qp}) & 0 \end{pmatrix} \tag{54}$$

Comparing Equations (53) and (54), $AJA^T = BJB^T$. Therefore, the Symplectic Euler method is exactly symplectic. □

5 The Störmer-Verlet Scheme

The Störmer-Verlet scheme is one of the most prominent numerical methods in physics, particularly designed for a specific class of second-order ordinary differential equations where the acceleration depends strictly on position:

$$\ddot{q} = f(q) \tag{55}$$

To analyze this geometrically, we transform the single second-order equation into a system of first-order equations. By defining an auxiliary velocity variable $v = \dot{q}$, we can substitute the original equation into its derivative to obtain the coupled system:

$$\dot{q} = v, \quad \dot{v} = f(q) \tag{56}$$

This decomposition directly mirrors the Hamiltonian formulation $\dot{q} = p, \dot{p} = f(q)$ for systems with normalized mass matrices.

5.1 The Classical 3-Term Recurrence and the Startup Problem

Historically, the Störmer-Verlet method was discretized by directly approximating the second derivative using a central difference formula. This yields the classical three-term recurrence relation:

$$q_{n+1} - 2q_n + q_{n-1} = h^2 f(q_n) \tag{57}$$

While computationally cheap because it does not require storing velocities, this recurrence suffers from a fundamental "startup problem". In physical mechanics, an initial value problem provides the initial position $q(0) = q_0$ and initial velocity $v(0) = v_0$. However, computing the very first step q_1 (at $n = 0$) requires the state q_{-1}:

$$q_1 - 2q_0 + q_{-1} = h^2 f(q_0) \tag{58}$$

The term q_{-1} represents a non-existent "ghost step" at time $t = -h$, meaning the classical method cannot begin on its own.

5.2 Deriving the Preliminary Starting Condition

To resolve the missing q_{-1} state, we utilize the given initial velocity v_0. We approximate v_0 using a central difference over the interval $[-h, h]$:

$$v_0 = \frac{q_1 - q_{-1}}{2h} \tag{59}$$

We can solve Equation (59) for the ghost step:

$$q_{-1} = q_1 - 2hv_0 \tag{60}$$

Substituting this expression back into the initial recurrence (Equation (58)) eliminates the unknown past state:

$$q_1 - 2q_0 + (q_1 - 2hv_0) = h^2 f(q_0) \tag{61}$$

Grouping the terms yields:

$$2q_1 - 2q_0 - 2hv_0 = h^2 f(q_0) \tag{62}$$

Dividing the entire equation by 2 and isolating q_1 provides the starting condition obtained from the central-difference approximation required to initiate the simulation:

$$q_1 = q_0 + hv_0 + \frac{h^2}{2} f(q_0) \tag{63}$$

5.3 The Explicit One-Step Formulation (Velocity Verlet)

The classical multi-step formulation poses practical issues beyond the startup calculation; primarily, it requires approximating velocities externally if they are needed to compute kinetic energy or check momentum conservation.

To bypass these issues, the method is rewritten as a true explicit one-step map, often called Velocity Verlet. By tracking an auxiliary velocity variable defined at the half-step, we propagate $(q_n, v_n) \mapsto (q_{n+1}, v_{n+1})$ directly:

$$v_{n+1/2} = v_n + \frac{h}{2} f(q_n) \tag{64}$$

$$q_{n+1} = q_n + hv_{n+1/2} \tag{65}$$

$$v_{n+1} = v_{n+1/2} + \frac{h}{2} f(q_{n+1}) \tag{66}$$

This formulation tracks both positions and velocities cleanly at the end of every time step and naturally accommodates variable step sizes.

6 The Adjoint of a Method and Symmetry

To mathematically prove the properties of the Störmer-Verlet scheme, we must first establish the concept of the adjoint of a numerical method. In classical physics, conservative systems exhibit perfect time-reversibility: if a system evolves forward in time by t and then backward by t, it returns exactly to its initial state. Mathematically, if φ_t represents the exact flow of an autonomous differential system $\dot{y} = f(y)$, it satisfies the property $\varphi_{-t}^{-1} = \varphi_t$, or equivalently, $\varphi_{-t} \circ \varphi_t = I$.

However, standard numerical integrators generally do not preserve this reversibility. For instance, executing a step of size h with the Explicit Euler method and then reversing it with a step of size $-h$ using the same formula does not return the system to its starting point. To formalize this behavior, we define the adjoint method.

Definition 6.1 (Adjoint Method)**.** The adjoint method Φ_h^* of a numerical method Φ_h is the inverse map of the original method executed with a reversed time step $-h$:

$$\Phi_h^* := \Phi_{-h}^{-1} \tag{67}$$

In practical terms, calculating a forward step $y_1 = \Phi_h^*(y_0)$ with the adjoint method is implicitly defined by running the original method backward from y_1 to recover the initial state: $\Phi_{-h}(y_1) = y_0$.

Example 6.2 (The Adjoint of the Symplectic Euler Method)**.** To illustrate how the adjoint mapping is explicitly calculated, let Φ_h represent the standard Symplectic Euler method:

$$p_1 = p_0 - h\frac{\partial H}{\partial q}(p_1, q_0) \tag{68}$$

$$q_1 = q_0 + h\frac{\partial H}{\partial p}(p_1, q_0) \tag{69}$$

To find its adjoint, we set up the inverse step Φ_{-h}. We apply the original method Φ to the state (q_1, p_1) with a step size of $-h$, requiring that the output is precisely the initial state (q_0, p_0). To do this, we perform the following substitutions in our original equations:

- Initial state: $(q_0, p_0) \longrightarrow (q_1, p_1)$
- Final state: $(q_1, p_1) \longrightarrow (q_0, p_0)$
- Step size: $h \longrightarrow -h$

Applying these substitutions yields the backward step equation:

$$p_0 = p_1 - (-h)\frac{\partial H}{\partial q}(p_0, q_1) \tag{70}$$

$$q_0 = q_1 + (-h)\frac{\partial H}{\partial p}(p_0, q_1) \tag{71}$$

To write this as an explicit one-step method propagating forward from (q_0, p_0) to (q_1, p_1), we simplify the double negatives and rearrange algebraically:

$$p_0 = p_1 + h\frac{\partial H}{\partial q}(p_0, q_1) \tag{72}$$

$$q_0 = q_1 - h\frac{\partial H}{\partial p}(p_0, q_1) \tag{73}$$

Isolating the target final state variables p_1 and q_1, we obtain the mapped system representing the adjoint method Φ_h^*:

$$p_1 = p_0 - h\frac{\partial H}{\partial q}(p_0, q_1) \tag{74}$$

$$q_1 = q_0 + h\frac{\partial H}{\partial p}(p_0, q_1) \tag{75}$$

The adjoint operator adheres to several clean algebraic rules. Taking the adjoint of an adjoint returns the original method, meaning $(\Phi_h^*)^* = \Phi_h$. Furthermore, the adjoint of a composed sequence of methods reverses their order, perfectly mirroring matrix transposition: $(\Phi_h \circ \Psi_h)^* = \Psi_h^* \circ \Phi_h^*$.

These properties allow us to rigorously define numerical symmetry.

Definition 6.3 (Symmetric Method)**.** A numerical method is called *symmetric* (or *self-adjoint*) if it is mathematically identical to its own adjoint:

$$\Phi_h^* = \Phi_h \tag{76}$$

While methods like Explicit Euler are not symmetric (its adjoint is the Implicit Euler method), integrators like the Störmer-Verlet scheme are strictly symmetric.

Theorem 6.4 (Order and Truncation Error of the Adjoint Method)**.** *Let Φ_h be a one-step method of order p with a local error expansion given by:*

$$\Phi_h(y_0) = \varphi_h(y_0) + C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) \tag{77}$$

where φ_h is the exact flow of the differential equation and $C(y_0)$ is the principal error coefficient. Then, the adjoint method Φ_h^* has the exact same order p, and its local error expansion is:

$$\Phi_h^*(y_0) = \varphi_h(y_0) + (-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) \tag{78}$$

Furthermore, if the method is symmetric ($\Phi_h = \Phi_h^*$), its maximal order is strictly even.

Proof. The proof relies on a geometric construction known as "back-projection". Let us start at an initial state y_0 and define two final points after a forward step of size h:

1. The exact flow point: $y_{\text{exact}} = \varphi_h(y_0)$.
2. The adjoint step point: $y_1 = \Phi_h^*(y_0)$.

The difference between these two points is the local truncation error of the adjoint method, which we denote as e^*:

$$e^* = y_1 - \varphi_h(y_0) \tag{79}$$

To evaluate this, we "project" both of these final points back to the starting time by applying the original numerical method with a negative step size, $-h$. By the fundamental definition of the adjoint method ($\Phi_h^* := \Phi_{-h}^{-1}$), running the negative step from the adjoint point y_1 lands us exactly back at our starting point:

$$\Phi_{-h}(y_1) = y_0 \tag{80}$$

Conversely, running the negative step from the exact flow point y_{exact} lands us at a back-projected point $\Phi_{-h}(\varphi_h(y_0))$. The difference between these two back-projected points defines the back-projected local error, e:

$$e = y_0 - \Phi_{-h}(\varphi_h(y_0)) \tag{81}$$

Because the exact flow of a differential equation is perfectly time-reversible, we know that $y_0 = \varphi_{-h}(\varphi_h(y_0))$. Substituting this into our expression for e yields:

$$e = \varphi_{-h}(\varphi_h(y_0)) - \Phi_{-h}(\varphi_h(y_0)) \tag{82}$$

Notice that e is exactly the negative of the local error of the original method Φ taking a negative step size $-h$, starting from the exact point $\overline{y} = \varphi_h(y_0)$. Using our known local error expansion formula for Φ_h, we substitute $-h$ for the step size:

$$e = -\left(\Phi_{-h}(\overline{y}) - \varphi_{-h}(\overline{y})\right) = -\left(C(\overline{y})(-h)^{p+1} + \mathcal{O}((-h)^{p+2})\right) \tag{83}$$

Factoring out the negative sign gives:

$$e = (-1)^p C(\varphi_h(y_0))h^{p+1} + \mathcal{O}(h^{p+2}) \tag{84}$$

Deriving the Jacobian of the Step Map

Every standard, consistent numerical one-step method $\Phi_h(y)$ can be expressed in terms of an increment function $\psi(y, h)$:

$$\Phi_h(y) = y + h\psi(y, h) \tag{85}$$

For the method to be consistent with the differential equation $\dot{y} = f(y)$, the increment function must satisfy $\psi(y, 0) = f(y)$. Since the vector field $f(y)$ is smooth, the increment function $\psi(y, h)$ is continuously differentiable (C^1) with respect to the state variable y.

To compute the Jacobian matrix of this step map, we take the partial derivative of $\Phi_h(y)$ with respect to y:

$$\frac{\partial \Phi_h(y)}{\partial y} = \frac{\partial}{\partial y}\left[y + h\psi(y, h)\right] = \frac{\partial y}{\partial y} + \frac{\partial}{\partial y}\left[h\psi(y, h)\right] \tag{86}$$

The derivative of the state vector with respect to itself is the identity matrix I. Factoring out the scalar step size h yields:

$$\frac{\partial \Phi_h(y)}{\partial y} = I + h\frac{\partial \psi(y, h)}{\partial y} \tag{87}$$

Because $\psi(y, h)$ is C^1, its Jacobian matrix $\frac{\partial \psi}{\partial y}$ is bounded in norm by some real constant M on any compact local neighbourhood of y:

$$\left\| \frac{\partial \psi(y, h)}{\partial y} \right\| \leq M \tag{88}$$

By the definition of asymptotic big-$\mathcal{O}$ notation, any matrix-valued function scaled by h whose norm is bounded by a constant times h is of order $\mathcal{O}(h)$:

$$h\frac{\partial \psi(y, h)}{\partial y} = \mathcal{O}(h) \tag{89}$$

Substituting this back into the Jacobian expression confirms that for any step size of order $\mathcal{O}(h)$ (including the negative step $-h$):

$$\frac{\partial \Phi_{-h}(y)}{\partial y} = I + \mathcal{O}(h) \tag{90}$$

Relating the Back-Projected Error e to the Forward Error e^*

Because the numerical step map Φ_{-h} is smooth and differentiable, we apply a multivariable Taylor expansion to relate the difference in its outputs to the difference in its inputs:

$$\Phi_{-h}(y_1) - \Phi_{-h}(\varphi_h(y_0)) = \frac{\partial \Phi_{-h}}{\partial y} \cdot (y_1 - \varphi_h(y_0)) + \mathcal{O}\left(\|y_1 - \varphi_h(y_0)\|^2\right) \tag{91}$$

Recalling the definition of our points, we have $\Phi_{-h}(y_1) = y_0$, which makes the left-hand side precisely our back-projected local error $e = y_0 - \Phi_{-h}(\varphi_h(y_0))$. Similarly, the term in parentheses on the right-hand side is our forward local error $e^* = y_1 - \varphi_h(y_0)$.

Substituting the Jacobian $\frac{\partial \Phi_{-h}}{\partial y} = I + \mathcal{O}(h)$ and noting that the quadratic error term $\mathcal{O}(\|e^*\|^2) = \mathcal{O}(h^{2p+2})$ is strictly absorbed into higher orders, we obtain:

$$e = (I + \mathcal{O}(h))e^* \tag{92}$$

Inverting this matrix relation yields:

$$e^* = (I + \mathcal{O}(h))^{-1} e = (I + \mathcal{O}(h))e \tag{93}$$

Evaluating the Principal Error Coefficient at y_0

From the previous stage of the proof, taking a negative step size $-h$ from the exact forward point $\varphi_h(y_0)$ yields a back-projected error of:

$$e = (-1)^p C(\varphi_h(y_0))h^{p+1} + \mathcal{O}(h^{p+2}) \tag{94}$$

However, we require the final formula for the adjoint error e^* to be evaluated at the initial state y_0. Because the exact flow over a time interval h moves a distance proportional to the step size, we have $\varphi_h(y_0) = y_0 + \mathcal{O}(h)$.

Since the principal error coefficient function $C(y)$ is continuous and smooth, we evaluate it at this shifted point via Taylor expansion:

$$C(\varphi_h(y_0)) = C(y_0 + \mathcal{O}(h)) = C(y_0) + \mathcal{O}(h) \tag{95}$$

Substituting this back into our expression for e gives:

$$e = (-1)^p \left[C(y_0) + \mathcal{O}(h)\right] h^{p+1} + \mathcal{O}(h^{p+2}) \tag{96}$$

Distributing the h^{p+1} factor:

$$e = (-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) + \mathcal{O}(h^{p+2}) = (-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2}) \tag{97}$$

Analyzing the Higher-Order Terms for e^*

Finally, we substitute this simplified expression for e back into our inverted Jacobian relation:

$$e^* = (I + \mathcal{O}(h)) \left[(-1)^p C(y_0)h^{p+1} + \mathcal{O}(h^{p+2})\right] \tag{98}$$

Multiplying this out term-by-term:

$$e^* = I \cdot \left[(-1)^p C(y_0)h^{p+1}\right] + I \cdot \mathcal{O}(h^{p+2}) + \mathcal{O}(h) \cdot \left[(-1)^p C(y_0)h^{p+1}\right] + \mathcal{O}(h) \cdot \mathcal{O}(h^{p+2}) \tag{99}$$

We analyze the order of each component as $h \to 0$:

1. **First Term:** The leading-order error contribution, $(-1)^p C(y_0)h^{p+1}$.
2. **Second Term:** The primary remainder of order $\mathcal{O}(h^{p+2})$.
3. **Third Term:** Since $\mathcal{O}(h) \cdot h^{p+1} = \mathcal{O}(h^{p+2})$, this term is of order $\mathcal{O}(h^{p+2})$.
4. **Fourth Term:** Since $\mathcal{O}(h) \cdot \mathcal{O}(h^{p+2}) = \mathcal{O}(h^{p+3})$, this term decays even faster.

Since the second, third, and fourth terms are all of order $\mathcal{O}(h^{p+2})$ or higher, they collapse into a single remainder bucket:

$$e^* = (-1)^p C(y_0) h^{p+1} + \mathcal{O}(h^{p+2}) \tag{100}$$

This perfectly matches the required local error expansion of the adjoint method, completing the first part of the proof. □

Corollary 6.5 (Even-Order Rule for Symmetric Methods)**.** *If a numerical method is symmetric, it is equal to its own adjoint: $\Phi_h = \Phi_h^*$. Consequently, their local truncation error expansions must be identical. Equating their leading-order coefficients yields:*

$$C(y_0) = (-1)^p C(y_0) \tag{101}$$

If the method's order p is odd, then $(-1)^p = -1$, which algebraically forces:

$$C(y_0) = -C(y_0) \implies 2C(y_0) = 0 \implies C(y_0) = 0 \tag{102}$$

Because the principal error coefficient $C(y_0)$ is forced to be exactly zero, the leading error term vanishes completely. The actual local error reduces to at least $\mathcal{O}(h^{p+2})$, which guarantees the physical order of accuracy jumps to at least $p+1$ (an even integer). Thus, the maximal order of a symmetric method must be even.

Theorem 6.6 (Symplecticity of the Adjoint Method)**.** *If a numerical mapping Φ_h is symplectic for any step size h, its adjoint method $\Phi_h^* = \Phi_{-h}^{-1}$ is also strictly symplectic.*

Proof. The proof leverages the algebraic group properties of symplectic matrices applied to the Jacobians of the mappings.

Step 1: Symplecticity of the Negative Step By definition, if the numerical method Φ_h is symplectic, its Jacobian matrix $G = \Phi_h'(y)$ satisfies the symplecticity condition for any step size h:

$$G^T J G = J \tag{103}$$

Since $-h$ is simply a negative step size, the negative-step map Φ_{-h} must also be a valid symplectic transformation:

$$(\Phi_{-h}')^T J (\Phi_{-h}') = J \tag{104}$$

Step 2: Symplecticity of the Inverse (The Jacobian Level) Let $M = \Phi_{-h}'(y)$ be the Jacobian of our negative-step map. Since it is symplectic, we have:

$$M^T J M = J \tag{105}$$

By the inverse function theorem, the Jacobian of the inverse map Φ_{-h}^{-1} (which represents our adjoint method Φ_h^*) is simply the matrix inverse M^{-1}. To prove the adjoint is symplectic, we must show that this inverse Jacobian is also symplectic:

$$(M^{-1})^T J M^{-1} = J \tag{106}$$

We start with our known symplecticity condition for the forward Jacobian:

$$M^T J M = J \tag{107}$$

Since M is invertible, we can multiply on the left by the inverse transpose $(M^T)^{-1}$ and on the right by the inverse M^{-1}:

$$(M^T)^{-1}(M^T J M)M^{-1} = (M^T)^{-1} J M^{-1} \tag{108}$$

Using the identity matrix properties on the left-hand side, the terms cancel out:

$$I \cdot J \cdot I = (M^T)^{-1} J M^{-1} \implies J = (M^T)^{-1} J M^{-1} \tag{109}$$

Since the matrix operations of transpose and inversion commute—meaning $(M^T)^{-1} = (M^{-1})^T$—we substitute this into our equation to get:

$$(M^{-1})^T J M^{-1} = J \tag{110}$$

This mathematically proves that the Jacobian of the inverse map M^{-1} is strictly symplectic. Consequently, the adjoint of any symplectic method is guaranteed to preserve symplecticity. □

7 Composition Methods

The composition method is an elegant and powerful technique designed to construct high-order numerical integrators from simple, low-order basic methods. A primary advantage of this approach is that if the underlying basic method preserves a specific geometric structure—such as symplecticity—the resulting composed method is mathematically guaranteed to preserve that exact same structure.

Suppose we are given a basic one-step numerical method Φ_h of order p. Instead of advancing the system by a single step of size h, we split the step into s fractional sub-steps of sizes $\gamma_1 h, \gamma_2 h, \ldots, \gamma_s h$. We apply the basic method sequentially using these sub-step sizes to define the corresponding composition method Ψ_h:

$$\Psi_h = \Phi_{\gamma_s h} \circ \cdots \circ \Phi_{\gamma_2 h} \circ \Phi_{\gamma_1 h} \tag{111}$$

Theorem 7.1 (Order-Increase Conditions for Composition Methods)**.** *Let Φ_h be a one-step method of order p. If the fractional step coefficients γ_i satisfy the consistency and order-increase conditions:*

$$\gamma_1 + \gamma_2 + \cdots + \gamma_s = 1 \tag{112}$$

$$\gamma_1^{p+1} + \gamma_2^{p+1} + \cdots + \gamma_s^{p+1} = 0 \tag{113}$$

then the composition method Ψ_h is at least of order $p+1$.

Proof. We analyze the global error of the composed method by tracking how the leading local truncation error accumulates and propagates across each of the s sub-steps.

Step 1: Defining the Local Stage Errors Let y_0 be the initial state and let y_{i-1} be the numerical state before the i-th sub-step. The local stage error e_i is the error committed strictly during the i-th sub-step of size $\gamma_i h$, defined as the difference between the numerical step and the exact flow $\varphi_{\gamma_i h}$ starting from the same intermediate point:

$$e_i = \Phi_{\gamma_i h}(y_{i-1}) - \varphi_{\gamma_i h}(y_{i-1}) \tag{114}$$

Because the basic method Φ is of order p, this local error is characterized by a principal error coefficient function $C(y)$:

$$e_i = C(y_{i-1})(\gamma_i h)^{p+1} + \mathcal{O}(h^{p+2}) \tag{115}$$

Step 2: Evaluating the Error Coefficient at y_0 Because each sub-step is of size $\mathcal{O}(h)$, the intermediate state variables remain close to the initial state, satisfying $y_{i-1} = y_0 + \mathcal{O}(h)$. Assuming the continuous error coefficient function $C(y)$ is smooth, we can expand it around y_0 using a Taylor series:

$$C(y_{i-1}) = C(y_0 + \mathcal{O}(h)) = C(y_0) + \mathcal{O}(h) \tag{116}$$

Substituting this expansion back into our expression for the local stage error yields:

$$e_i = (C(y_0) + \mathcal{O}(h))\gamma_i^{p+1} h^{p+1} + \mathcal{O}(h^{p+2}) = C(y_0)\gamma_i^{p+1} h^{p+1} + \mathcal{O}(h^{p+2}) \tag{117}$$

Step 3: Propagating Stage Errors to the Final Time The local error e_i committed at stage i does not remain static; it propagates under the exact flow of the differential equation for the remainder of the full time step. Let E_i denote this propagated error evaluated at the final time $t = h$.

As established in the analysis of flow maps, the exact flow over a small time interval behaves as the identity mapping plus a perturbation of order $\mathcal{O}(h)$. Consequently, the Jacobian matrix that propagates this small discrepancy forward is $I + \mathcal{O}(h)$. Therefore, the propagated stage error relates to the local stage error by:

$$E_i = (I + \mathcal{O}(h))e_i \tag{118}$$

Step 4: Summing the Telescoping Errors The total global error of the composed step is the difference between the exact flow over the full step and the numerical composed step. By expressing this as a telescoping sum of the individual propagated stage errors (Lady Windermere's Fan), we obtain:

$$\varphi_h(y_0) - \Psi_h(y_0) = \sum_{i=1}^{s} E_i = \sum_{i=1}^{s} (I + \mathcal{O}(h))e_i \tag{119}$$

Substituting our expanded expression for e_i into the sum:

$$\varphi_h(y_0) - \Psi_h(y_0) = \sum_{i=1}^{s} (I + \mathcal{O}(h)) \left(C(y_0)\gamma_i^{p+1} h^{p+1} + \mathcal{O}(h^{p+2}) \right) \tag{120}$$

Multiplying this out, any term containing $\mathcal{O}(h) \cdot h^{p+1}$ collapses into the $\mathcal{O}(h^{p+2})$ remainder bucket. We can then factor out the common leading-order components:

$$\varphi_h(y_0) - \Psi_h(y_0) = C(y_0) \left(\sum_{i=1}^{s} \gamma_i^{p+1} \right) h^{p+1} + \mathcal{O}(h^{p+2}) \tag{121}$$

Conclusion By enforcing the algebraic condition of the theorem, $\sum_{i=1}^{s} \gamma_i^{p+1} = 0$, the entire leading error term strictly vanishes. Consequently, the total error of the composed method Ψ_h reduces to:

$$\varphi_h(y_0) - \Psi_h(y_0) = \mathcal{O}(h^{p+2}) \tag{122}$$

This mathematically proves that the composed method achieves an order of at least $p + 1$. □

Theorem 7.2 (Symplecticity of Composition Methods)**.** *Let Φ_h be a symplectic numerical method. Then, any composition method defined by*

$$\Psi_h = \Phi_{\gamma_s h} \circ \cdots \circ \Phi_{\gamma_1 h} \tag{123}$$

remains strictly symplectic for any arbitrary choice of real step coefficients $\gamma_1, \ldots, \gamma_s$.

Proof. The proof relies on the algebraic property that symplectic transformations form a mathematical group under function composition. We demonstrate this by tracking the Jacobian of composed mappings.

Step 1: The Definition of a Symplectic Mapping Recall that a differentiable mapping $g : U \to \mathbb{R}^{2d}$ is symplectic if and only if its Jacobian matrix $g'(y)$ satisfies the symplecticity condition everywhere on its domain:

$$g'(y)^T J g'(y) = J \tag{124}$$

where J is the standard skew-symmetric structure matrix.

Step 2: Composition of Two Symplectic Mappings Let g_1 and g_2 be two arbitrary symplectic mappings. We evaluate the symplecticity of their composition $G = g_2 \circ g_1$. By the chain rule of multivariable calculus, the Jacobian of the composed map $G(y) = g_2(g_1(y))$ is the product of their individual Jacobians:

$$G'(y) = g_2'(g_1(y)) \cdot g_1'(y) \tag{125}$$

We test if this composed Jacobian satisfies the definition of symplecticity:

$$G'(y)^T J G'(y) = \left(g_2'(g_1(y)) g_1'(y)\right)^T J \left(g_2'(g_1(y)) g_1'(y)\right) \tag{126}$$

Using the matrix transpose identity $(AB)^T = B^T A^T$, we expand the right-hand side:

$$G'(y)^T J G'(y) = g_1'(y)^T \left[g_2'(g_1(y))^T J g_2'(g_1(y)) \right] g_1'(y) \tag{127}$$

Because g_2 is a symplectic mapping, its Jacobian satisfies $g_2'(z)^T J g_2'(z) = J$ for any point z in its domain. Substituting $z = g_1(y)$ collapses the inner bracketed term directly to J:

$$G'(y)^T J G'(y) = g_1'(y)^T J g_1'(y) \tag{128}$$

Similarly, because g_1 is also symplectic, its Jacobian satisfies $g_1'(y)^T J g_1'(y) = J$. Substituting this yields:

$$G'(y)^T J G'(y) = J \tag{129}$$

This rigorously proves that the composition of two symplectic mappings is always strictly symplectic.

Step 3: Mathematical Induction By induction, this algebraic property generalizes to the composition of any finite sequence of s symplectic mappings. If $g_1, g_2, \ldots, g_s$ are all individual symplectic transformations, then their sequential composition $G = g_s \circ \cdots \circ g_2 \circ g_1$ is guaranteed to satisfy $G'(y)^T J G'(y) = J$, making the final composed mapping strictly symplectic.

Step 4: Application to the Composition Method In a general composition method, we define our composed one-step map Ψ_h by sequentially applying our basic method Φ_h using a set of step coefficients:

$$\Psi_h = \Phi_{\gamma_s h} \circ \cdots \circ \Phi_{\gamma_1 h} \tag{130}$$

Because the basic method Φ_h is inherently symplectic, each individual sub-stage map $g_i = \Phi_{\gamma_i h}$ is a perfectly valid symplectic mapping (it is simply the basic symplectic method evaluated with an individual step size of $h_i = \gamma_i h$). Since each sub-stage is symplectic, their composition Ψ_h must be strictly and unconditionally symplectic. □

Composition with the Adjoint Method

In our standard composition method, we encountered a fundamental algebraic bottleneck: if the basic method's order p is odd, the condition $\sum \gamma_i^{p+1} = 0$ has no real, non-zero solutions because $p+1$ is even, making every term strictly positive. To achieve an order increase for methods with odd p, we must generalize our approach by explicitly interleaving the basic method Φ_h with its adjoint Φ_h^*.

We replace the standard composition formula with a more general sequence utilizing two sets of step coefficients, α_i and β_i:

$$\Psi_h = \Phi_{\alpha_s h} \circ \Phi^*_{\beta_s h} \circ \cdots \circ \Phi_{\alpha_2 h} \circ \Phi^*_{\beta_2 h} \circ \Phi_{\alpha_1 h} \circ \Phi^*_{\beta_1 h} \tag{131}$$

To determine the conditions required for this composed method Ψ_h to achieve order $p+1$, we rely on Theorem 6.4, which establishes that the leading local error term of the adjoint method is scaled by $(-1)^p$. By accumulating the propagated stage errors as we did previously, the condition for order $p+1$ becomes:

$$\beta_1 + \alpha_1 + \beta_2 + \alpha_2 + \cdots + \beta_s + \alpha_s = 1 \tag{132}$$

$$(-1)^p \beta_1^{p+1} + \alpha_1^{p+1} + (-1)^p \beta_2^{p+1} + \alpha_2^{p+1} + \cdots + (-1)^p \beta_s^{p+1} + \alpha_s^{p+1} = 0 \tag{133}$$

Because of the alternating $(-1)^p$ factor introduced by the adjoint steps, the error terms can successfully cancel one another out even when p is odd, entirely resolving the previous real-solution bottleneck.

The $s = 1$ Symmetric Construction

A particularly powerful and immediate application of this generalized formula occurs for a single stage ($s = 1$) starting from a consistent, first-order method ($p = 1$). The order conditions simplify to:

$$\beta_1 + \alpha_1 = 1 \tag{134}$$

$$-\beta_1^2 + \alpha_1^2 = 0 \tag{135}$$

We can see at once that the solution is $\alpha_1 = \beta_1 = 1/2$. This implies that every consistent one-step method of order 1 can be automatically transformed into a second-order, symmetric method via the composition:

$$\Psi_h = \Phi_{h/2} \circ \Phi^*_{h/2} \tag{136}$$

8 The General Störmer-Verlet Method for Hamiltonian Systems

Initially, we discussed the Störmer-Verlet method in the context of a simple second-order differential equation. However, for an arbitrary, fully coupled, non-separable Hamiltonian $H(p, q)$, the Störmer-Verlet method is formally defined precisely as the half-step composition of the Symplectic Euler method with its adjoint:

$$\Psi_h = \Phi_{h/2} \circ \Phi^*_{h/2} \tag{137}$$

Without making any assumptions about the structure of $H(p, q)$, this composition yields a 4-step implicit algorithm to advance the system from $(q_n, p_n) \to (q_{n+1}, p_{n+1})$.

Step 1: The Adjoint Half-Step $\Phi^*_{h/2}$ Using the definition of the adjoint of Symplectic Euler, we solve the implicit equations for the intermediate coordinates $(q_{n+1/2}, p_{n+1/2})$:

$$q_{n+1/2} = q_n + \frac{h}{2} \frac{\partial H}{\partial p}(p_n, q_{n+1/2}) \tag{138}$$

$$p_{n+1/2} = p_n - \frac{h}{2} \frac{\partial H}{\partial q}(p_n, q_{n+1/2}) \tag{139}$$

Step 2: The Symplectic Euler Half-Step $\Phi_{h/2}$ Using the standard Symplectic Euler equations, we complete the step to reach (q_{n+1}, p_{n+1}):

$$p_{n+1} = p_{n+1/2} - \frac{h}{2}\frac{\partial H}{\partial q}(p_{n+1}, q_{n+1/2}) \tag{140}$$

$$q_{n+1} = q_{n+1/2} + \frac{h}{2}\frac{\partial H}{\partial p}(p_{n+1}, q_{n+1/2}) \tag{141}$$

The Explicit Formulation for Separable Hamiltonians

For a single stage ($s = 1$) starting from a first-order method ($p = 1$), the consistency and order conditions yield a symmetric composition with half-step sizes: $\Psi_h = \Phi_{h/2} \circ \Phi^*_{h/2}$. We can prove algebraically that this composition corresponds exactly to the canonical Störmer-Verlet scheme for a separable physical Hamiltonian of the form $H(p, q) = T(p) + U(q)$.

Step 1: Apply the Adjoint Method with step size $h/2$ Starting from (q_n, p_n), we compute the intermediate half-step state $(q_{n+1/2}, p_{n+1/2}) = \Phi^*_{h/2}(q_n, p_n)$:

$$q_{n+1/2} = q_n + \frac{h}{2}\frac{\partial H}{\partial p}(p_n, q_{n+1/2}) = q_n + \frac{h}{2}\nabla T(p_n) \tag{142}$$

$$p_{n+1/2} = p_n - \frac{h}{2}\frac{\partial H}{\partial q}(p_n, q_{n+1/2}) = p_n - \frac{h}{2}\nabla U(q_{n+1/2}) \tag{143}$$

Step 2: Apply the Original Method with step size $h/2$ Starting from $(q_{n+1/2}, p_{n+1/2})$, we complete the full step $(q_{n+1}, p_{n+1}) = \Phi_{h/2}(q_{n+1/2}, p_{n+1/2})$:

$$p_{n+1} = p_{n+1/2} - \frac{h}{2}\frac{\partial H}{\partial q}(p_{n+1}, q_{n+1/2}) = p_{n+1/2} - \frac{h}{2}\nabla U(q_{n+1/2}) \tag{144}$$

$$q_{n+1} = q_{n+1/2} + \frac{h}{2}\frac{\partial H}{\partial p}(p_{n+1}, q_{n+1/2}) = q_{n+1/2} + \frac{h}{2}\nabla T(p_{n+1}) \tag{145}$$

Step 3: Eliminating the Intermediate Momentum $p_{n+1/2}$ Substituting the equation for $p_{n+1/2}$ directly into the update equation for p_{n+1} yields:

$$p_{n+1} = \left[p_n - \frac{h}{2}\nabla U(q_{n+1/2})\right] - \frac{h}{2}\nabla U(q_{n+1/2}) = p_n - h\nabla U(q_{n+1/2}) \tag{146}$$

Gathering these steps in sequence gives the nice explicit **Position-Verlet** formulation of the Störmer-Verlet scheme:

$$q_{n+1/2} = q_n + \frac{h}{2}\nabla T(p_n) \tag{147}$$

$$p_{n+1} = p_n - h\nabla U(q_{n+1/2}) \tag{148}$$

$$q_{n+1} = q_{n+1/2} + \frac{h}{2}\nabla T(p_{n+1}) \tag{149}$$

If we compose the methods in the reverse order ($\Psi_h = \Phi^*_{h/2} \circ \Phi_{h/2}$), we recover the standard explicit **Velocity-Verlet (Momentum-Verlet)** formulation:

$$p_{n+1/2} = p_n - \frac{h}{2}\nabla U(q_n) \tag{150}$$

$$q_{n+1} = q_n + h\nabla T(p_{n+1/2}) \tag{151}$$

$$p_{n+1} = p_{n+1/2} - \frac{h}{2}\nabla U(q_{n+1}) \tag{152}$$

Theorem 8.1 (Symplecticity and Order of the Störmer-Verlet Scheme)**.** *The Störmer-Verlet scheme is a symplectic numerical method of order 2.*

Proof. We prove this using the mathematical properties of composition and adjoint mappings established previously in this report.

Part A: Proof of Symplecticity

1. **Symplecticity of the Base Method:** By Theorem 4.2, the Symplectic Euler method Φ_h is a symplectic map for any smooth Hamiltonian system.

2. **Symplecticity of the Adjoint:** By Theorem 6.6, since the adjoint of any symplectic map is also symplectic, the Adjoint Symplectic Euler method Φ_h^* is unconditionally symplectic.

3. **Symplecticity of Composition:** By Theorem 7.2, any composition method of the form $\Psi_h = \Phi_{\gamma_s h} \circ \cdots \circ \Phi_{\gamma_1 h}$, or any generalized composition utilizing the adjoint, is strictly symplectic for any choice of step coefficients if all the functions being composed are symplectic.

4. **Conclusion of Symplecticity:** Since the Störmer-Verlet scheme is defined precisely as the composition of two symplectic mappings, namely:

$$\Psi_h = \Phi_{h/2} \circ \Phi_{h/2}^* \tag{153}$$

   it is mathematically guaranteed to be strictly symplectic.

Part B: Proof of Order 2 (Symmetry)

1. **The Transpose Rule of Composed Adjoints:** For any composed sequence, the adjoint operator reverses the order of composition:

$$(\Phi_h \circ \Psi_h)^* = \Psi_h^* \circ \Phi_h^* \tag{154}$$

2. **Symmetry of Störmer-Verlet:** We evaluate the adjoint of the complete Störmer-Verlet map Ψ_h:

$$\Psi_h^* = \left(\Phi_{h/2} \circ \Phi_{h/2}^*\right)^* = (\Phi_{h/2}^*)^* \circ \Phi_{h/2}^* \tag{155}$$

   Using the double adjoint property, $(\Phi^*)^* = \Phi$, this expression simplifies to:

$$\Psi_h^* = \Phi_{h/2} \circ \Phi_{h/2}^* = \Psi_h \tag{156}$$

   Since $\Psi_h^* = \Psi_h$, the Störmer-Verlet scheme is strictly symmetric (self-adjoint).

3. **The Even-Order Rule:** By Theorem 6.4 and Corollary 6.5, the maximal order of accuracy p of any symmetric method must be strictly even.

4. Since the base method is consistent (meaning $p \geq 1$), and the symmetry constraint analytically forces an even order, the order of the Störmer-Verlet method automatically jumps to order 2.

This completes the rigorous, structure-preserving proof of Theorem 8.1. □

9 Relaxation Runge-Kutta Methods for Hamiltonian Systems

9.1 Theoretical Foundations of Relaxation Runge-Kutta (RRK) Methods

Standard explicit and diagonally implicit Runge-Kutta methods do not conserve nonlinear invariants. Unmodified explicit schemes cause numerical trajectories to drift across energy level curves, leading to unphysical phase-space expansion or dissipation.

Relaxation Runge-Kutta (RRK) methods, introduced by Ketcheson [2], modify standard Runge-Kutta schemes by introducing a scalar relaxation parameter $\gamma_n \approx 1$ at each time step. Given an s-stage baseline Runge-Kutta method, the internal stages Y_i and the baseline update direction vector d_n are computed normally:

$$Y_i = y_n + \Delta t \sum_{j=1}^{s} a_{ij} f(Y_j), \quad i = 1, \ldots, s \tag{157}$$

$$d_n = \sum_{i=1}^{s} b_i f(Y_i) \tag{158}$$

In the RRK framework, the update direction d_n is preserved, but the step length is dynamically scaled by γ_n:

$$y_{n+1}^{(\gamma)} = y_n + \gamma_n \Delta t d_n \tag{159}$$

The scalar relaxation factor γ_n is uniquely determined at step n by enforcing exact conservation of the target invariant $H_0 = H(y_0)$:

$$G(\gamma_n) \equiv H\left(y_n + \gamma_n \Delta t d_n\right) - H_0 = 0 \tag{160}$$

9.2 Formulation and Properties of Relaxed Heun's Method

A. Continuous Pendulum System

As established, the normalized mathematical pendulum is governed by the Hamiltonian:

$$H(q,p) = \frac{1}{2}p^2 - \cos(q) \tag{161}$$

Applying Hamilton's canonical equations yields the vector field $f(y)$:

$$\dot{y} = f(y) \implies \begin{pmatrix} \dot{q} \\ \dot{p} \end{pmatrix} = \begin{pmatrix} p \\ -\sin(q) \end{pmatrix} \tag{162}$$

B. Baseline Heun's Method (Explicit RK2)

For a state $y_n = (q_n, p_n)^T$ and step size h, Heun's method evaluates two stage derivatives:

$$k_1 = f(y_n) = \begin{pmatrix} p_n \\ -\sin(q_n) \end{pmatrix} \tag{163}$$

$$k_2 = f(y_n + hk_1) = \begin{pmatrix} p_n - h\sin(q_n) \\ -\sin(q_n + hp_n) \end{pmatrix} \tag{164}$$

The baseline update direction vector d_n is constructed from the arithmetic average of these two stages:

$$d_n = \begin{pmatrix} d_n^q \\ d_n^p \end{pmatrix} = \frac{1}{2}\left(k_1 + k_2\right) \tag{165}$$

C. The Extra Relaxation Step and Newton-Raphson Mechanics

Instead of taking the standard explicit step $y_{n+1} = y_n + hd_n$ (which inherently causes artificial energy drift), the relaxation approach scales the step length by a scalar parameter $\gamma_n \approx 1$:

$$y_{n+1}^{(\gamma)} = \begin{pmatrix} q_{n+1}^{(\gamma)} \\ p_{n+1}^{(\gamma)} \end{pmatrix} = y_n + \gamma_n h d_n = \begin{pmatrix} q_n + \gamma_n h d_n^q \\ p_n + \gamma_n h d_n^p \end{pmatrix} \tag{166}$$

The scalar parameter γ_n is uniquely determined at each time step by enforcing exact conservation of the target Hamiltonian, ensuring the energy matches the global reference energy H_0:

$$G(\gamma_n) \equiv H\left(y_{n+1}^{(\gamma)}\right) - H_0 = 0 \tag{167}$$

For the pendulum, this residual equation expands to:

$$\frac{1}{2}\left(p_n + \gamma_n h d_n^p\right)^2 - \cos\left(q_n + \gamma_n h d_n^q\right) - H_0 = 0 \tag{168}$$

Because $G(\gamma)$ is a smooth scalar algebraic equation, γ_n can be computed efficiently at every step using a 1D Newton-Raphson iteration starting from an initial guess of $\gamma^{(0)} = 1.0$:

$$\gamma^{(k+1)} = \gamma^{(k)} - \frac{G(\gamma^{(k)})}{G'(\gamma^{(k)})} \tag{169}$$

The derivative $G'(\gamma)$ required for the iteration is obtained via the multivariable chain rule as the directional derivative of the Hamiltonian along the update direction d_n:

$$G'(\gamma) = \nabla H\left(y_n + \gamma h d_n\right)^T \cdot (hd_n) = h\left[\sin\left(q_n + \gamma h d_n^q\right) d_n^q + \left(p_n + \gamma h d_n^p\right) d_n^p\right] \tag{170}$$

To render the relaxation scheme completely robust across all regions of phase space—particularly near singularities where the vector alignment between the directional derivative d_n and the energy gradient ∇H becomes orthogonal ($G'(\gamma) \to 0$)—three mandatory numerical safeguards are built into the root-finding routine:

1. **Global Reference Energy:** The target energy H_0 is evaluated once globally at $t = 0$ to prevent local energy drift.
2. **Derivative Cutoff:** If $|G'(\gamma)| < 10^{-14}$, the iteration terminates and sets $\gamma = 1.0$ for that step.
3. **Parameter Bounds:** The Newton correction step is clamped to $|\Delta\gamma| \leq 0.5$, and γ_n is strictly bounded within $[0.2, 1.8]$.

10 Higher Order Symplectic Methods

To construct and analyze higher-order symplectic integrators, it is highly advantageous to examine the evolution of the system's variational equations. We will formally establish that for any numerical integrator, preserving the quadratic invariant of the variational equation is strictly equivalent to the method being symplectic.

10.1 The Variational Equation and Quadratic Invariants

Consider the $2d$-dimensional autonomous Hamiltonian system defined on a phase space domain $U \subset \mathbb{R}^{2d}$, which we previously expressed as $\dot{y} = J^{-1}\nabla H(y)$. Differentiating the exact flow with respect to the initial conditions y_0 yields the matrix variational system:

$$\dot{\Psi}(t) = J^{-1}\nabla^2 H(y(t))\,\Psi(t), \quad \Psi(0) = \Psi_0 \tag{171}$$

where $\nabla^2 H(y)$ is the symmetric Hessian matrix of the Hamiltonian.

Lemma 10.1 (The Continuous Quadratic Invariant)**.** *For any solution pair $(y(t), \Psi(t))$ of the variational system, the quadratic form $Q(\Psi) := \Psi^T J \Psi$ is a first integral (a constant of motion).*

Proof. Differentiating $Q(\Psi(t)) = \Psi(t)^T J \Psi(t)$ with respect to time along the trajectories yields:

$$\frac{d}{dt}\left(\Psi^T J \Psi\right) = \dot{\Psi}^T J \Psi + \Psi^T J \dot{\Psi} \tag{172}$$

Substituting the variational equation $\dot{\Psi} = J^{-1} S \Psi$, where $S(t) = \nabla^2 H(y(t))$ is symmetric ($S^T = S$):

$$\dot{\Psi}^T J \Psi = \left(J^{-1} S \Psi\right)^T J \Psi = \Psi^T S^T (J^{-1})^T J \Psi = \Psi^T S(-J^{-1}) J \Psi = -\Psi^T S \Psi \tag{173}$$

Similarly, for the second term:

$$\Psi^T J \dot{\Psi} = \Psi^T J \left(J^{-1} S \Psi\right) = \Psi^T S \Psi \tag{174}$$

Adding these two evaluated terms gives:

$$\frac{d}{dt}\left(\Psi^T J \Psi\right) = -\Psi^T S \Psi + \Psi^T S \Psi = 0 \tag{175}$$

Thus, the quadratic form is exactly conserved continuously. □

Lemma 10.2 (Commutativity of Discretization and Variational Differentiation)**.** *For a numerical one-step method applied to $\dot{y} = f(y)$, the numerical solution sequence $\{\Psi_n\}$ obtained by applying the method to the augmented variational system $\dot{\Psi} = f'(y)\Psi$ with initial condition $\Psi_0 = I$ is identically equal to the Jacobian matrix of the discrete numerical step map:*

$$\Psi_n = \frac{\partial y_n}{\partial y_0} = \frac{\partial \Phi_h^n(y_0)}{\partial y_0} \tag{176}$$

Proof. Consider a general one-step Runge-Kutta map. Differentiating the discrete state update equations with respect to the initial state y_0 via the multivariable chain rule yields the exact algebraic equations produced by applying the same Runge-Kutta scheme to the linear system $\dot{\Psi} = f'(y)\Psi$. Since the initial condition is $\Psi_0 = \frac{\partial y_0}{\partial y_0} = I$, mathematical induction establishes that $\Psi_n = \frac{\partial y_n}{\partial y_0}$ holds identically for all steps $n \geq 0$. □

10.2 Equivalence Theorem

Building upon the geometric definition of a symplectic numerical map (Definition 4.1), we can now prove the fundamental equivalence theorem for numerical integrators.

Theorem 10.3 (Equivalence of Symplecticity and Quadratic Invariant Preservation)**.** *A numerical integrator $y_{n+1} = \Phi_h(y_n)$ is symplectic if and only if its numerical step preserves the quadratic form $Q(\Psi) = \Psi^T J \Psi$ for the associated variational system.*

Proof.

($\implies$ Direction: Symplecticity implies Quadratic Invariant Preservation) Assume that the numerical one-step map $y_{n+1} = \Phi_h(y_n)$ is symplectic. As established in Section 2, its step Jacobian matrix $G(y_n) := \frac{\partial \Phi_h(y_n)}{\partial y_n}$ satisfies the geometric condition:

$$G(y_n)^T J G(y_n) = J \tag{177}$$

Now apply the numerical method to the linear variational system $\dot{\Psi} = J^{-1}\nabla^2 H(y)\Psi$ starting from an arbitrary initial matrix $\Psi_n \in \mathbb{R}^{2d\times 2d}$. By the linearity of the variational equation and Lemma 10.2, advancing the variational state by one step produces:

$$\Psi_{n+1} = G(y_n)\,\Psi_n \tag{178}$$

Evaluating the quadratic form $Q(\Psi_{n+1})$ at step $n+1$:

$$Q(\Psi_{n+1}) = \Psi_{n+1}^T J \Psi_{n+1} = \left(G(y_n)\Psi_n\right)^T J \left(G(y_n)\Psi_n\right) = \Psi_n^T \left(G(y_n)^T J G(y_n)\right) \Psi_n \tag{179}$$

Substituting the symplecticity property $G(y_n)^T J G(y_n) = J$:

$$Q(\Psi_{n+1}) = \Psi_n^T J \Psi_n = Q(\Psi_n) \tag{180}$$

Therefore, the numerical method strictly preserves the quadratic invariant $Q(\Psi) = \Psi^T J \Psi$ across every time step.

($\impliedby$ Direction: Preservation of Quadratic Invariant implies Symplecticity) Assume that the numerical method preserves the quadratic invariant $Q(\Psi) = \Psi^T J \Psi$ when applied to the variational system. That is, for any step $\Psi_n \to \Psi_{n+1}$, the method guarantees:

$$\Psi_{n+1}^T J \Psi_{n+1} = \Psi_n^T J \Psi_n \tag{181}$$

Select the initial variational matrix to be the identity matrix at step n, setting $\Psi_n = I_{2d}$. Evaluating the quadratic form at step n gives:

$$\Psi_n^T J \Psi_n = I^T J I = J \tag{182}$$

By Lemma 10.2, when $\Psi_n = I$, the numerical update for the variational matrix after one forward step is identically the step Jacobian $G(y_n) = \frac{\partial \Phi_h(y_n)}{\partial y_n}$:

$$\Psi_{n+1} = G(y_n) \tag{183}$$

Substituting $\Psi_{n+1} = G(y_n)$ into the quadratic invariant preservation condition $\Psi_{n+1}^T J \Psi_{n+1} = J$ yields:

$$G(y_n)^T J G(y_n) = J \tag{184}$$

Since this relation holds for every state $y_n \in U$, the Jacobian $G(y_n)$ is everywhere symplectic by definition. Thus, the numerical method is rigorously proven to be symplectic. □

10.3 Collocation Methods

To systematically construct high-order implicit Runge-Kutta methods—particularly those that satisfy the geometric conditions for symplecticity—we utilize the framework of polynomial collocation. This approach constructs a continuous polynomial trajectory $u(t)$ that satisfies the differential equation exactly at s distinct nodal points.

The profound connection between collocation and Runge-Kutta methods is established by the following theorem, which demonstrates that finding the continuous collocation polynomial is algebraically identical to executing an implicit Runge-Kutta step.

Theorem 10.4 (Guillou & Soulé 1969, Wright 1970)**.** *The collocation method based on s distinct nodes $c_1, \ldots, c_s$ is mathematically equivalent to an s-stage Runge-Kutta method with coefficients defined by the integrals of Lagrange interpolating polynomials:*

$$a_{ij} = \int_0^{c_i} \ell_j(\tau)\, d\tau, \quad b_i = \int_0^1 \ell_i(\tau)\, d\tau \tag{185}$$

where $\ell_i(\tau)$ is the standard Lagrange basis polynomial defined as:

$$\ell_i(\tau) = \prod_{l \neq i} \frac{\tau - c_l}{c_i - c_l} \tag{186}$$

Proof. Let $u(t)$ be the collocation polynomial of degree $\leq s$. Because its time derivative $\dot{u}(t)$ is a polynomial of degree $\leq s-1$, it is uniquely determined by its values at the s distinct node points. We define these derivative values as the Runge-Kutta stage derivatives k_i:

$$k_i := \dot{u}(t_0 + c_i h) \tag{187}$$

Using the normalized time variable $\tau = \frac{t - t_0}{h}$, we apply the Lagrange interpolation formula to express the derivative polynomial exactly everywhere as a linear combination of the basis polynomials:

$$\dot{u}(t_0 + \tau h) = \sum_{j=1}^{s} k_j \ell_j(\tau) \tag{188}$$

To determine the internal state of the system at the i-th collocation node, we integrate this derivative from the starting time t_0 ($\tau = 0$) to the node time $t_0 + c_i h$ ($\tau = c_i$):

$$u(t_0 + c_i h) = y_0 + h \sum_{j=1}^{s} k_j \int_0^{c_i} \ell_j(\tau)\, d\tau \tag{189}$$

By substituting the definition $a_{ij} = \int_0^{c_i} \ell_j(\tau)\, d\tau$, this expression perfectly matches the first formula defining the internal stages of an s-stage Runge-Kutta equation.

To obtain the final step update $y_1 = u(t_0 + h)$, we integrate the derivative polynomial over the full normalized time step from $\tau = 0$ to $\tau = 1$. This integration yields the second formula of the Runge-Kutta equation, confirming that $b_i = \int_0^1 \ell_i(\tau)\, d\tau$. □

10.4 Gauss Collocation Methods and Interpolatory Quadrature

Interpolatory and Gaussian Quadrature

An s-point interpolatory quadrature rule is a general numerical integration formula built by replacing an integrand $g(\tau)$ over a normalized domain $\tau \in [0, 1]$ with its degree-$(s-1)$ Lagrange interpolating polynomial passing through s arbitrary distinct node points $c_1, c_2, \ldots, c_s \in [0, 1]$. The quadrature weights b_i are defined as the exact integrals of the Lagrange basis polynomials $\ell_i(\tau)$ over $[0, 1]$:

$$b_i = \int_0^1 \ell_i(\tau)\, d\tau, \quad \text{where } \ell_i(\tau) = \prod_{k \neq i} \frac{\tau - c_k}{c_i - c_k} \tag{190}$$

Because the weights are derived from Lagrange basis polynomials, the rule integrates every monomial τ^{k-1} for $k = 1, \ldots, s$ exactly. Thus, a general s-point interpolatory rule works for any set of distinct nodes c_i and is guaranteed to be exact for all polynomials up to degree $s-1$. As a result, the rule has a baseline precision order of at least $p = s$.

The s-point Gaussian Quadrature Rule is a specific, highly optimized interpolatory rule used to approximate the definite integral of a function over an interval using s discrete evaluation points. The s quadrature nodes $c_1, c_2, \ldots, c_s \in (0, 1)$ are chosen as the s distinct roots of the s-th shifted Legendre polynomial $P_s(x)$ on $[0, 1]$:

$$P_s(x) = \frac{d^s}{dx^s}\left(x^s (x-1)^s\right) \tag{191}$$

While a standard interpolatory rule with s points integrates polynomials up to degree $s-1$ exactly, choosing the nodes c_i as the roots of $P_s(x)$ boosts the precision to order $p = 2s$. That is, the rule integrates any polynomial $g(\tau)$ of degree up to $2s-1$ with zero numerical error:

$$\int_0^1 g(\tau)\, d\tau = \sum_{i=1}^{s} b_i\, g(c_i) \quad \forall\, g \in \mathcal{P}_{2s-1} \tag{192}$$

This superconvergence relies on polynomial division and the orthogonality of Legendre polynomials. Any polynomial $g(\tau)$ of degree $\leq 2s-1$ can be decomposed via long division as $g(\tau) = Q(\tau)M(\tau) + R(\tau)$, where $M(\tau) = \prod_{i=1}^{s}(\tau - c_i)$ is proportional to the shifted Legendre polynomial, and both the quotient $Q(\tau)$ and remainder $R(\tau)$ have degrees $\leq s-1$. Because $Q(\tau)$ has degree $\leq s-1$, it is orthogonal to $M(\tau)$, meaning $\int_0^1 Q(\tau)\, M(\tau)\, d\tau = 0$. The exact integral thus reduces strictly to the integral of the remainder: $\int_0^1 g(\tau)\, d\tau = \int_0^1 R(\tau)\, d\tau$. When applying the s-point quadrature sum, the quotient term vanishes at every node because $M(c_i) = 0$. Since the remainder $R(\tau)$ has degree $\leq s-1$, the interpolatory rule integrates it exactly. Thus, the quadrature sum perfectly evaluates the continuous integral.

Conservation of Quadratic Invariants

The proof that Gauss collocation methods conserve all quadratic invariants is one of the most elegant proofs in geometric numerical integration. It hinges on a single key property: s-stage Gaussian quadrature integrates polynomials up to degree $2s-1$ with zero error.

Consider a general autonomous differential system $\dot{y} = f(y)$ with $y(t_0) = y_0$. A quadratic function $Q(y) = y^T C y$ (where C is a symmetric matrix) is a first integral (invariant) of $\dot{y} = f(y)$ if its directional derivative vanishes along all trajectory solutions:

$$\frac{d}{dt}Q(y(t)) = 2y(t)^T C\dot{y}(t) = 2y(t)^T C f(y(t)) = 0 \quad \forall y \tag{193}$$

Thus, the condition for $Q(y) = y^T C y$ to be an invariant is that $y^T C f(y) = 0$ holds for every state y.

Given s distinct nodes $c_1, c_2, \ldots, c_s \in [0,1]$ chosen as the s roots of the shifted Legendre polynomial $P_s(x) = \frac{d^s}{dx^s}\left(x^s(x-1)^s\right)$, the collocation polynomial $u(t)$ is a vector polynomial of degree at most s satisfying:

1. Initial condition: $u(t_0) = y_0$.
2. Collocation conditions at the s node times $t_0 + c_i h$: $\dot{u}(t_0 + c_i h) = f(u(t_0 + c_i h))$ for $i = 1, \ldots, s$.

The discrete numerical update after one step h is defined as $y_1 = u(t_0 + h)$.

Theorem 10.5 (Hairer, Wanner & Lubich [1])**.** *The Gauss collocation methods based on the shifted Legendre polynomials conserve all quadratic invariants $Q(y) = y^T C y$ exactly: $Q(y_1) = Q(y_0)$.*

Proof. By the Fundamental Theorem of Calculus, the net change in the quadratic invariant $Q(y)$ over the time step $[t_0, t_0 + h]$ along the collocation polynomial trajectory $u(t)$ is:

$$Q(y_1) - Q(y_0) = Q(u(t_0 + h)) - Q(u(t_0)) = \int_{t_0}^{t_0+h} \frac{d}{dt}Q(u(t))\, dt \tag{194}$$

Differentiating $Q(u(t)) = u(t)^T C u(t)$ using the product rule gives:

$$\frac{d}{dt}Q(u(t)) = \dot{u}(t)^T C u(t) + u(t)^T C\dot{u}(t) = 2u(t)^T C\dot{u}(t) \tag{195}$$

Substituting this back into the integral yields:

$$y_1^T C y_1 - y_0^T C y_0 = \int_{t_0}^{t_0+h} 2u(t)^T C\dot{u}(t)\, dt \tag{196}$$

We now examine the degree of the integrand polynomial $P(t) := u(t)^T C\dot{u}(t)$. Because $u(t)$ is a polynomial of degree $\leq s$, its derivative $\dot{u}(t)$ is a polynomial of degree $\leq s-1$. Therefore, the scalar product $u(t)^T C\dot{u}(t)$ is a polynomial in t of degree at most:

$$\text{degree}(u) + \text{degree}(\dot{u}) \leq s + (s-1) = 2s - 1 \tag{197}$$

An s-point Gaussian quadrature rule with nodes $c_1, \ldots, c_s$ and positive weights $b_1, \ldots, b_s$ integrates any polynomial of degree up to $2s-1$ with zero numerical error. Because $2u(t)^T C\dot{u}(t)$ has degree $\leq 2s-1$, the continuous integral over $[t_0, t_0+h]$ is identically equal to its s-point Gauss quadrature sum:

$$\int_{t_0}^{t_0+h} 2u(t)^T C\dot{u}(t)\,dt = h\sum_{i=1}^{s} b_i \cdot \Big(2u(t_0+c_ih)^T C\dot{u}(t_0+c_ih)\Big) \tag{198}$$

At each Gauss node time $t_0 + c_ih$, the collocation polynomial satisfies the exact ODE derivative condition $\dot{u}(t_0+c_ih) = f(u(t_0+c_ih))$. Replacing $\dot{u}(t_0+c_ih)$ with $f(u(t_0+c_ih))$ inside the quadrature sum gives:

$$\int_{t_0}^{t_0+h} 2u(t)^T C\dot{u}(t)\,dt = 2h\sum_{i=1}^{s} b_i \cdot \Big(u(t_0+c_ih)^T Cf(u(t_0+c_ih))\Big) \tag{199}$$

Recall the definition of a quadratic invariant: $y^T Cf(y) = 0$ holds for all vectors y. Evaluating this identity at the specific state vectors $y = u(t_0+c_ih)$ means:

$$u(t_0+c_ih)^T Cf(u(t_0+c_ih)) = 0 \quad \text{for each } i = 1, 2, \ldots, s \tag{200}$$

Consequently, every single term in the Gaussian quadrature sum is identically zero:

$$\int_{t_0}^{t_0+h} 2u(t)^T C\dot{u}(t)\,dt = 2h\sum_{i=1}^{s} b_i \cdot (0) = 0 \tag{201}$$

Substituting this result back into our initial integral relation gives:

$$y_1^T Cy_1 - y_0^T Cy_0 = 0 \implies Q(y_1) = Q(y_0) \tag{202}$$

Thus, the numerical state y_1 preserves the quadratic invariant $Q(y)$ to machine precision. □

Theorem 10.6 (Cooper 1987)**.** *If the coefficients of a Runge-Kutta method satisfy*

$$b_ia_{ij} + b_ja_{ji} = b_ib_j \quad \textit{for all } i, j = 1, \ldots, s, \tag{203}$$

then it conserves quadratic invariants.

Proof. Let us start with the numerical update relation $y_1 = y_0 + h\sum_{i=1}^{s} b_ik_i$. Evaluating the quadratic form $y_1^T Cy_1$ gives:

$$y_1^T Cy_1 = y_0^T Cy_0 + h\sum_{i=1}^{s} b_ik_i^T Cy_0 + h\sum_{j=1}^{s} b_jy_0^T Ck_j + h^2\sum_{i,j=1}^{s} b_ib_jk_i^T Ck_j \tag{204}$$

We can write the internal stage derivatives as $k_i = f(Y_i)$, where the stage values are defined by $Y_i = y_0 + h\sum_{j=1}^{s} a_{ij}k_j$. The central idea is to compute y_0 from this relation, $y_0 = Y_i - h\sum_{j=1}^{s} a_{ij}k_j$, and insert it directly into the cross-terms of our expanded equation. Using the symmetry of C, this substitution yields:

$$y_1^T Cy_1 = y_0^T Cy_0 + 2h\sum_{i=1}^{s} b_iY_i^T Cf(Y_i) + h^2\sum_{i,j=1}^{s} (b_ib_j - b_ia_{ij} - b_ja_{ji})k_i^T Ck_j \tag{205}$$

By the core assumption that $y^T Cy$ is an invariant of the differential system, we are guaranteed that $Y_i^T Cf(Y_i) = 0$ holds for every stage i. The required algebraic condition $b_ia_{ij} + b_ja_{ji} = b_ib_j$ forces the terms in the h^2 summation to identically cancel out. Consequently, we are left with $y_1^T Cy_1 = y_0^T Cy_0$, completing the proof that the method conserves the quadratic invariant perfectly. □

10.5 From Skew-Symmetric Bilinear Forms to Symmetric Quadratic Invariants

A crucial caveat arises when connecting Cooper's theorem to symplecticity: Cooper's condition (Theorem 10.6) assumes that the invariant matrix C is *symmetric* ($C^T = C$). In contrast, the canonical symplectic structure matrix J governing the differential 2-form $\omega(\xi, \eta) = \xi^T J\eta$ is fundamentally *skew-symmetric* ($J^T = -J$).

To bridge this distinction and apply Cooper's theorem directly to prove symplecticity, we reformulate the preservation of the skew-symmetric bilinear form between any two variational vectors $\xi, \eta \in \mathbb{R}^{2d}$ as the preservation of a symmetric quadratic form on an extended phase space $\mathbb{R}^{4d}$.

1. Constructing the Extended State and Symmetric Matrix C

We stack the two independent tangent (variational) vectors $\xi, \eta \in \mathbb{R}^{2d}$ into a single extended state vector $Y \in \mathbb{R}^{4d}$:

$$Y = \begin{pmatrix} \xi \\ \eta \end{pmatrix} \in \mathbb{R}^{4d} \tag{206}$$

Next, we define the block $4d \times 4d$ matrix C:

$$C = \frac{1}{2} \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix} \tag{207}$$

Taking the transpose of C and using the skew-symmetry of J ($J^T = -J$):

$$C^T = \frac{1}{2} \begin{pmatrix} 0 & (-J)^T \\ J^T & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & -(-J) \\ -J & 0 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix} = C \tag{208}$$

Because J is skew-symmetric, the constructed block matrix C is strictly symmetric ($C^T = C$).

2. Evaluating the Quadratic Form $Y^T C Y$

We now evaluate the quadratic form $Q(Y) = Y^T C Y$ on $\mathbb{R}^{4d}$:

$$Y^T C Y = \frac{1}{2} \begin{pmatrix} \xi^T & \eta^T \end{pmatrix} \begin{pmatrix} 0 & J \\ -J & 0 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \frac{1}{2} \left(\xi^T J \eta - \eta^T J \xi \right) \tag{209}$$

Because $\eta^T J \xi$ is a scalar, it is identical to its transpose. Applying $J^T = -J$ yields:

$$\eta^T J \xi = (\eta^T J \xi)^T = \xi^T J^T \eta = -\xi^T J \eta$$

Substituting this relation back into the quadratic form gives:

$$Y^T C Y = \frac{1}{2} \left(\xi^T J \eta - (-\xi^T J \eta) \right) = \xi^T J \eta \tag{210}$$

3. Connection to Cooper's Theorem

Equation (210) establishes that the skew-symmetric symplectic bilinear form $\xi^T J \eta$ on $\mathbb{R}^{2d}$ is identically equal to the symmetric quadratic form $Y^T C Y$ on $\mathbb{R}^{4d}$.

Since C is symmetric, Cooper's theorem (Theorem 10.6) applies directly to $Q(Y) = Y^T C Y$. Therefore, any Runge-Kutta method whose coefficients satisfy Cooper's algebraic condition:

$$b_i a_{ij} + b_j a_{ji} = b_i b_j \quad \text{for all } i, j = 1, \ldots, s$$

conserves the quadratic invariant $Y^T C Y$, which is equivalent to preserving the symplectic form $\xi^T J \eta$ between any two variational solutions. By the Equivalence Theorem (Theorem 10.3), this guarantees that the method is strictly symplectic.

10.6 The 2-Stage Gauss-Legendre Method (Gauss 4)

To apply the theoretical framework developed in the preceding sections, we construct the 4th-order symplectic Runge-Kutta method derived directly from Gauss collocation. By selecting $s = 2$ stages and choosing the quadrature nodes as the roots of the shifted Legendre polynomial, we obtain the 2-stage Gauss-Legendre method (often referred to as the Hammer-Hollingsworth method or Gauss 4).

Because this method is an s-stage Gauss collocation scheme, it inherits all the powerful geometric properties previously established: it conserves all quadratic invariants exactly, is strictly symplectic (preserving 100% of the phase-space area), and achieves a superconvergent global order of accuracy $p = 2s = 4$.

The Butcher Tableau

The method is uniquely defined by the following Butcher tableau coefficients:

$$\begin{array}{c|cc} c_1 = \frac{1}{2} - \frac{\sqrt{3}}{6} & a_{11} = \frac{1}{4} & a_{12} = \frac{1}{4} - \frac{\sqrt{3}}{6} \\ c_2 = \frac{1}{2} + \frac{\sqrt{3}}{6} & a_{21} = \frac{1}{4} + \frac{\sqrt{3}}{6} & a_{22} = \frac{1}{4} \\ \hline & b_1 = \frac{1}{2} & b_2 = \frac{1}{2} \end{array} \tag{211}$$

It is straightforward to algebraically verify that these coefficients satisfy Cooper's condition (203), $b_i a_{ij} + b_j a_{ji} = b_i b_j$ for all $i, j \in \{1, 2\}$. By Theorem 10.6 and the skew-to-symmetric equivalence established in Section 10.5, satisfying this condition provides an independent algebraic guarantee of the method's exact symplecticity.

Update Equations for Implementation

To advance an autonomous differential system $\dot{y} = f(y)$ from a state y_n at time t_n to a new state y_{n+1} at time $t_n + h$, the method requires solving a fully coupled 2×2 block implicit system for the internal stage derivatives k_1 and k_2:

$$k_1 = f\left(y_n + h\left[\frac{1}{4}k_1 + \left(\frac{1}{4} - \frac{\sqrt{3}}{6}\right)k_2\right]\right) \tag{212}$$

$$k_2 = f\left(y_n + h\left[\left(\frac{1}{4} + \frac{\sqrt{3}}{6}\right)k_1 + \frac{1}{4}k_2\right]\right) \tag{213}$$

Once this nonlinear implicit system is solved (typically via a multidimensional Newton-Raphson iteration) for a given step, the final state update is computed as a simple arithmetic average of the evaluated stages:

$$y_{n+1} = y_n + \frac{h}{2}(k_1 + k_2) \tag{214}$$

11 Numerical Experiments

The simulation routines and MATLAB implementation scripts developed for all numerical experiments in this report are openly available in the accompanying project repository [3].

11.1 Experiment 1: Phase Space Area Evolution of Symplectic and Non-Symplectic Methods

To empirically demonstrate the failure of standard numerical integrators to preserve the geometric structure of Hamiltonian systems, we track the evolution of a defined phase space area over time.

The background trajectories for the system are generated by selecting multiple initial states along the p-axis and evolving them until $t = 15$ using a step size of $h = 0.5$ (0.05 for the explicit euler plot). These trajectories visually define the underlying flow of the differential equation.

We define an initial rectangular region in the phase space spanning the position coordinates $q \in [-1, 1]$ and the momentum coordinates $p \in [0.5, 1.5]$. The exact initial oriented area of this region is 2.0000. To approximate the continuous manifold computationally, we discretize the boundary of this rectangle by placing 500 equidistant points along each of its four edges, forming a closed, 2000-sided polygon.

Using this 2000-point boundary as our set of initial conditions, we propagate the system forward in time using four distinct, non-symplectic Runge-Kutta methods: Explicit Euler (order 1), Heun's Method (order 2), Standard RK3 (order 3), and Standard RK4 (order 4). We record the deformed boundary coordinates at $t = 2$ and $t = 5$. By applying the standard polygon area formula to these advected boundary points, we approximate the area of the deformed region. Specifically, the enclosed area is computed using the Gauss Shoelace formula (executed via MATLAB's `polyarea` function), which efficiently calculates the absolute area by summing the 2D cross products of adjacent vertex vectors:

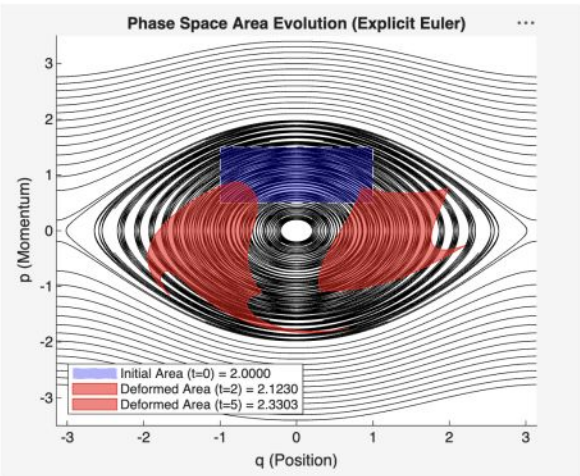
$$\text{Area} = \frac{1}{2}\left|\sum_{i=1}^{N}(q_i p_{i+1} - q_{i+1}p_i)\right| = \frac{1}{2}\left|\sum_{i=1}^{N}\det\begin{pmatrix} q_i & q_{i+1} \\ p_i & p_{i+1}\end{pmatrix}\right| \tag{215}$$

This calculation inherently assumes that the topological mapping is continuous, meaning that all points initially contained within the boundary remain strictly enclosed by the deformed boundary.

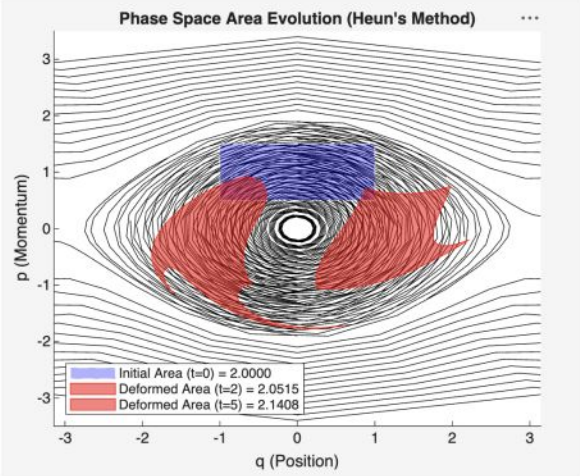
The experimental results explicitly confirm that non-symplectic methods do not conserve phase space area:

- Under the **Explicit Euler** method, the area artificially expands to 2.1230 at $t = 2$ and 2.3303 at $t = 5$.
- Under **Heun's Method**, the area expands to 2.0515 at $t = 2$ and 2.1408 at $t = 5$.
- Under **Standard RK3**, the area artificially dissipates to 1.9815 at $t = 2$ and 1.9515 at $t = 5$.
- Under **Standard RK4**, the area dissipates slightly to 1.9991 at $t = 2$ and 1.9976 at $t = 5$.

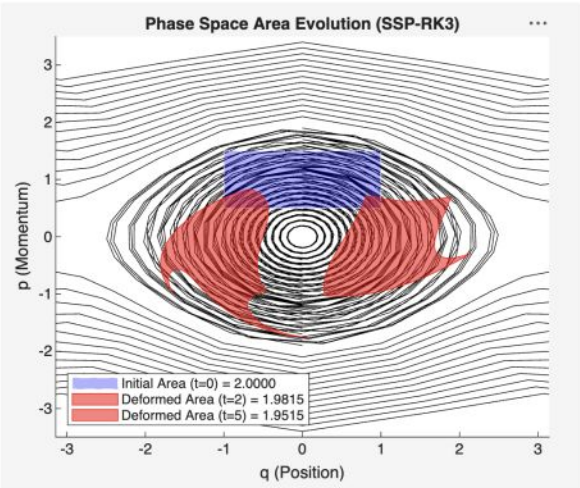
While none of these methods conserve the geometric area exactly, the magnitude of the area deviation from the true initial area monotonically decreases as the formal order of the numerical method increases.



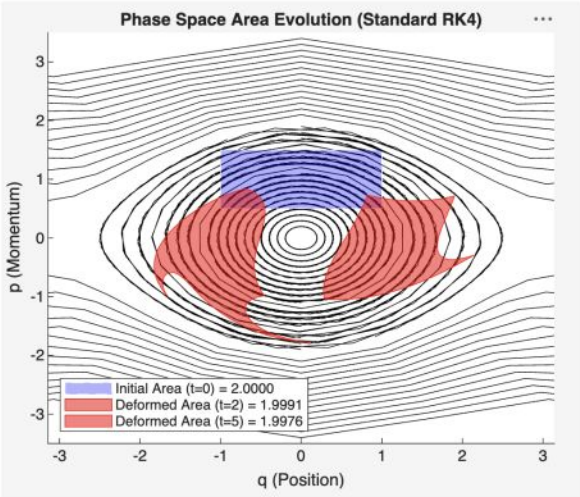
(a) Explicit Euler (Order 1)



(b) Heun's Method (Order 2)



(c) SSP-RK3 (Order 3)



(d) Standard RK4 (Order 4)

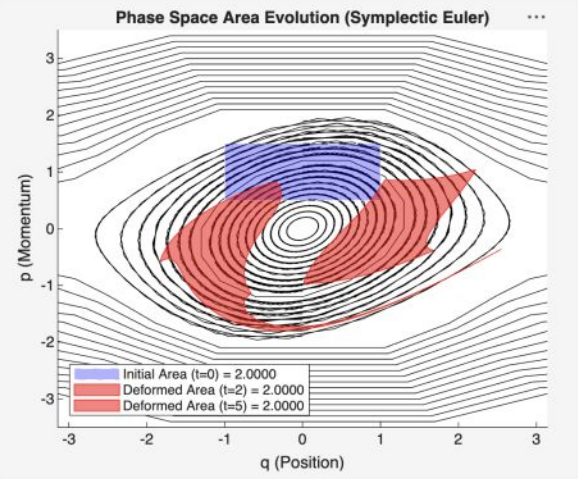
Figure 1: Phase space area evolution of a rectangular initial domain under various non-symplectic methods at $t = 0$, $t = 2$, and $t = 5$.

To contrast this with structure-preserving algorithms, we repeat the same experimental setup—an initial rectangular domain of nominal area 2.0000 discretized by 2000 boundary points, integrated with the step size $h = 0.5$—using two symplectic integrators: the Symplectic Euler method (order 1) and the Störmer-Verlet scheme (order 2).

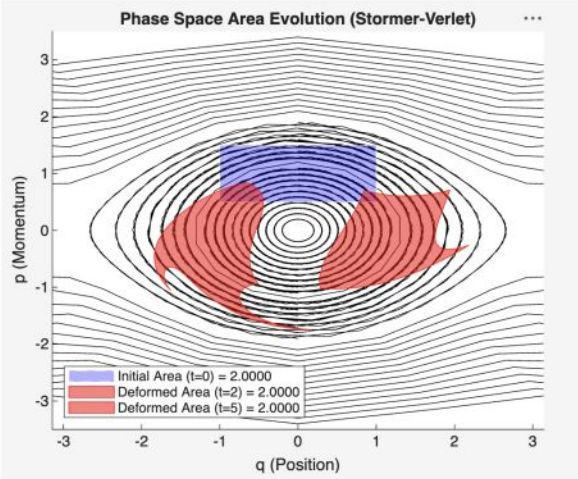
Because we are using a relatively large step size, there are noticeable truncation errors in the placement and deformation of the macroscopic shapes compared to the continuous flow of the differential equation. The "tails" of the advected regions stretch and warp over time.

However, the defining characteristic of these methods is their geometric structure preservation. The underlying numerical maps preserve area in exact arithmetic, while the polygon-based numerical estimates are subject to discretization and floating-point errors. For both methods, the calculated area of the advected 2000-sided polygon rounds to 2.0000 at $t = 2$ and $t = 5$.

This empirical result is consistent with the theoretical results established in the preceding sections: symplectic integrators preserve the geometric area of the phase space in exact arithmetic, independent of the chosen step size or the method's formal order of accuracy.



(a) Symplectic Euler (Order 1)



(b) Störmer-Verlet (Order 2)

Figure 2: Phase space area evolution under symplectic numerical methods at $t = 0$, $t = 2$, and $t = 5$. Despite the shape deformation caused by the large step size ($h = 0.5$), the computed oriented area remains 2.0000 (to four decimal places) over the simulation.

11.2 Experiment 2: Long-Term Area Conservation Analysis

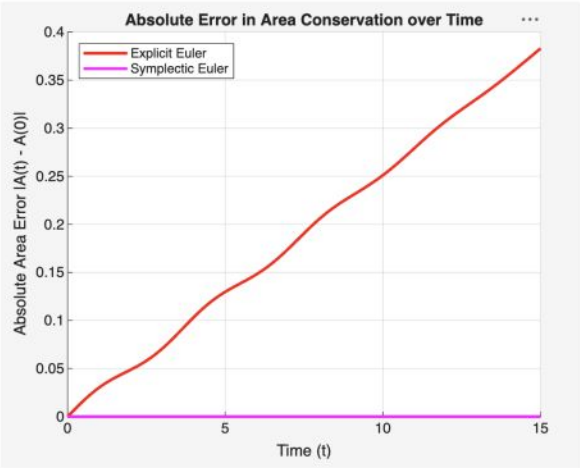
In our second experiment, we evaluate the structural stability of these numerical integrators by tracking the absolute error in the phase space area over time.

The initial rectangular domain remains identical to the first experiment. However, to ensure high precision over the integration periods, the boundary is discretized more finely by placing 1000 equidistant points along each edge, resulting in a total of 4000 boundary points. To provide a rigorous geometric evaluation, we compare methods of equivalent formal order. Because the parameters have been optimized for each comparison, we evaluate the first-order methods over a focused time domain and the second-order methods over an extended time domain.

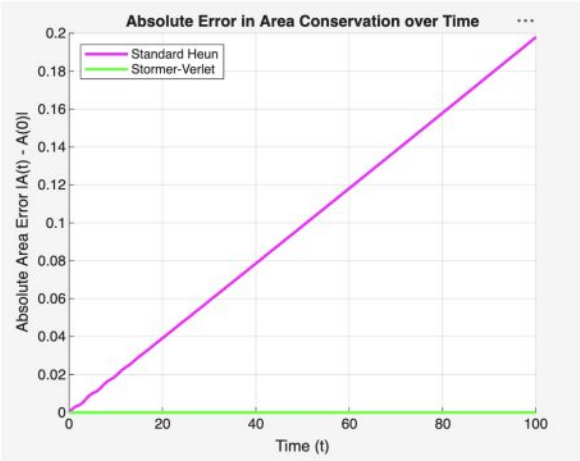
The resulting absolute error plots demonstrate a clear difference in behavior between the non-symplectic and symplectic integrators:

- **First-Order Comparison ($t \in [0, 15]$, $h = 0.02$):** The absolute area error of the non-symplectic Explicit Euler method steadily accumulates in a near-linear fashion, reaching an error magnitude close to 0.4 by the end of the simulation. Conversely, the Symplectic Euler method preserves the phase-space area, maintaining an absolute error close to zero over the interval.
- **Second-Order Comparison ($t \in [0, 100]$, $h = 0.2$):** Over the extended time domain, the higher-order non-symplectic Heun's Method exhibited increasing area error over the tested interval. In contrast, the symplectic Störmer-Verlet scheme conserved the phase-space volume, with its absolute error remaining flat near zero for the duration of the simulation.

These long-term experiments highlight a fundamental principle of geometric numerical integration: while non-symplectic methods can suffer from artificial energy dissipation or accumulation (manifesting as drifting phase space area), symplectic methods respect the underlying Hamiltonian structure globally, yielding bounded geometric errors over the simulated interval.



(a) Order 1: Explicit vs. Symplectic Euler



(b) Order 2: Heun's Method vs. Störmer-Verlet

Figure 3: Absolute error in phase space area conservation. The Order 1 comparison is evaluated over $t \in [0, 15]$ with $h = 0.02$, while the Order 2 comparison spans an extended time domain $t \in [0, 100]$ with $h = 0.2$. In both cases, symplectic methods maintain near-zero absolute error over the simulated interval, while their non-symplectic counterparts display increasing error growth over the tested interval.

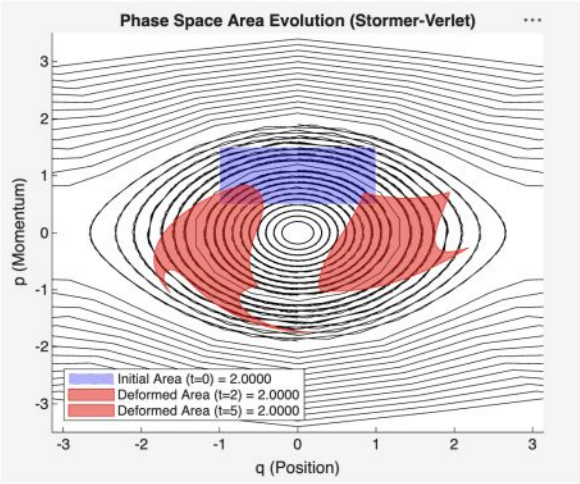
11.3 Experiment 3: Phase Space Area Evolution (Relaxed Heun vs. Störmer-Verlet)

To evaluate the phase-space area preservation properties of Heun's Method with Relaxation against the benchmark Störmer-Verlet Scheme, we construct a boundary advection experiment similar to Experiment 1.

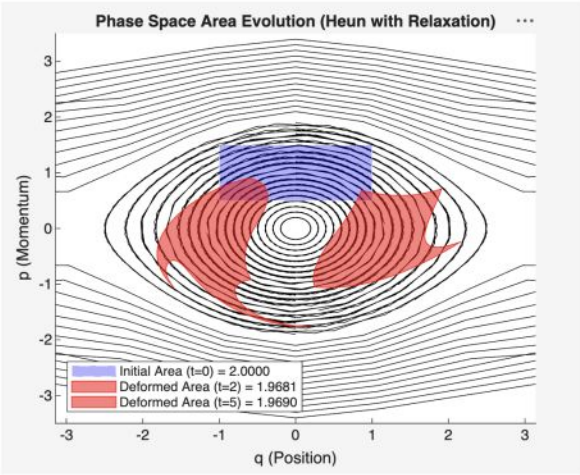
We define an initial rectangular domain in phase space spanning position coordinates $q \in [-1, 1]$ and momentum coordinates $p \in [0.5, 1.5]$, yielding an exact initial oriented area of 2.0000. To capture the geometric deformations with high precision, the boundary is discretized by placing 2000 equidistant points along each of its four edges, forming a closed, 8000-sided polygon. Each boundary point is independently integrated from $t = 0$ to the evaluation timestamps $t = 2$ and $t = 5$ using a nominal step size of $h = 0.5$. The area enclosed by the advected polygon is calculated using Gauss's shoelace formula.

The comparative results reveal the fundamental geometric difference between preserving the area in phase space versus preserving the physical Hamiltonian:

- **Störmer-Verlet Scheme (Symplectic):** Because this method algebraically satisfies the symplecticity condition at every step, the underlying numerical map preserves phase-space area in exact arithmetic, while the polygon-based numerical estimate is subject to discretization and floating-point errors. Despite the visible shape deformation and tail stretching caused by the large step size $h = 0.5$, the computed enclosed area rounds to 2.0000 at both $t = 2$ and $t = 5$.
- **Relaxed Heun Scheme (Energy-Preserving):** This method enforces exact energy conservation $H(y_k(t)) = H(y_k(0))$ for every individual boundary point k. Because each point experiences a distinct sequence of relaxation parameters γ_k depending on its local trajectory, the mapping is non-canonical ($\det J \neq 1$). This causes the polygon area to deform slightly to 1.9681 at $t = 2$ and 1.9690 at $t = 5$. Crucially, because the points are constrained to their physical energy level curves, this area does not expand monotonically like the standard baseline Heun's method; instead, it remained bounded over the simulated interval.



(a) Störmer-Verlet (Order 2)



(b) Heun with Relaxation (Order 2)

Figure 4: Phase space area evolution comparing a symplectic integrator to an energy-preserving Relaxation integrator at $t = 0$, $t = 2$, and $t = 5$ with $h = 0.5$. The Störmer-Verlet scheme preserves area in exact arithmetic (with computed values rounding to 2.0000), while the Relaxed Heun scheme allows bounded geometric deformation over the simulated interval while conserving physical energy.

11.4 Experiment 4: Comprehensive Comparison of Area and Energy Conservation (Order 2)

To illustrate the structure-preserving dichotomy between symplectic and energy-preserving integrators, we evaluate the long-term conservation properties of three second-order methods: the non-symplectic Standard Heun method, the symplectic Störmer-Verlet scheme, and the energy-preserving Relaxed Heun scheme. This experiment employs a dual-benchmark approach to measure both absolute area error and absolute energy error on both linear and logarithmic scales.

For the geometric area benchmark, we utilize the identical initial rectangular domain from the previous experiments ($q \in [-1, 1], p \in [0.5, 1.5]$ with an initial area of 2.0000). To ensure high resolution over the integration period, the boundary is discretized into 5000 points per edge (20,000 total boundary points), and the system is integrated from $t = 0$ to $t = 25$ using a step size of $h = 0.5$.

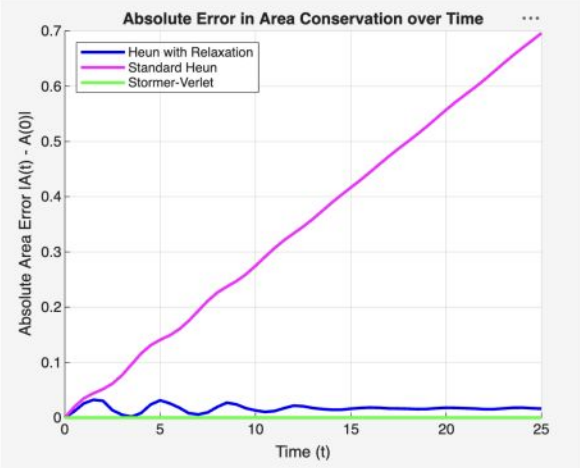
For the physical energy benchmark, we trace a single state trajectory initialized at $q_0 = 0.0$ and $p_0 = 1.0$ (corresponding to a true energy of $H_{\text{true}} = -0.5000$). This single trajectory is integrated over an extended time domain from $t = 0$ to $t = 200$ using a finer step size of $h = 0.05$, tracking the absolute energy error $|H(t) - H(0)|$.

The analytical synthesis of these long-term numerical results directly visualizes the theoretical theorems governing Hamiltonian integration:

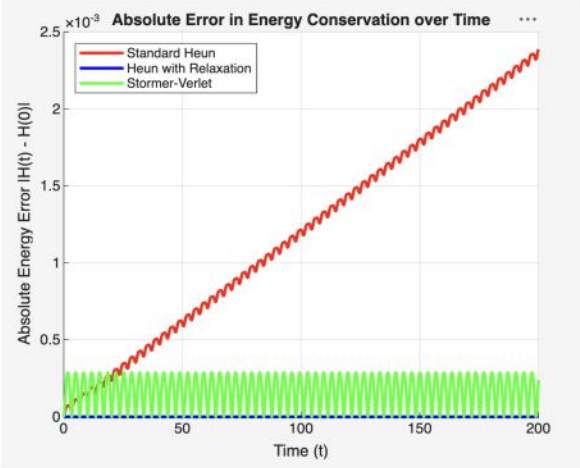
- **Area Conservation Dynamics:** The non-symplectic Standard Heun method exhibited increasing area error over the tested interval, approaching an absolute error of 0.7 by $t = 25$. In contrast, the energy-preserving Relaxed Heun method experiences an initial transient geometric deformation, but its area error oscillates and remained bounded below 0.035 over the simulated interval, preventing secular phase-space expansion over this duration. The symplectic Störmer-Verlet scheme theoretically preserves the phase-space area in exact arithmetic; while its linear plot appears flat at zero, the logarithmic scale reveals a residual error of order 10^{-6}. This minute deviation is not a failure of the symplectic method, but rather a geometric artifact of using the discrete `polyarea` function, which inherently suffers from truncation error when approximating the continuous curved manifold with a finite 20,000-sided polygon.

- **Energy Conservation Dynamics:** The Standard Heun method demonstrates increasing secular energy drift over the tested interval. Conversely, the absolute error of the symplectic Störmer-Verlet method oscillates rapidly within a bounded envelope. The Relaxed Heun scheme enforces energy preservation at each step; on the logarithmic scale, its energy error drops near machine precision (fluctuating between 10^{-16} and 10^{-12}). These fluctuations represent the numerical noise introduced by the Newton-Raphson solver's finite tolerance when computing the nonlinear relaxation parameter γ.

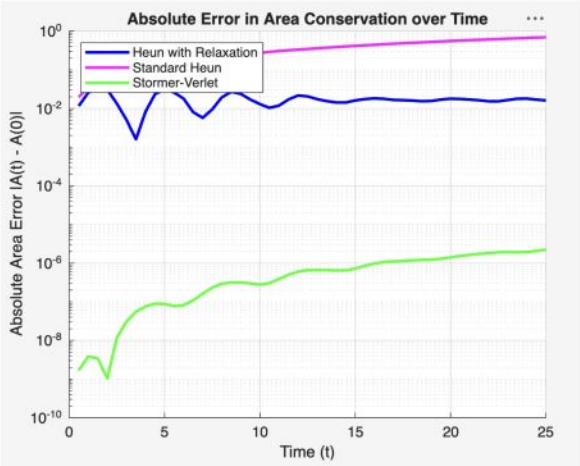
These experiments show that the methods considered do not generally preserve both the Hamiltonian and phase-space area exactly. However, they can improve the long-term behaviour of quantities that they do not preserve by construction.



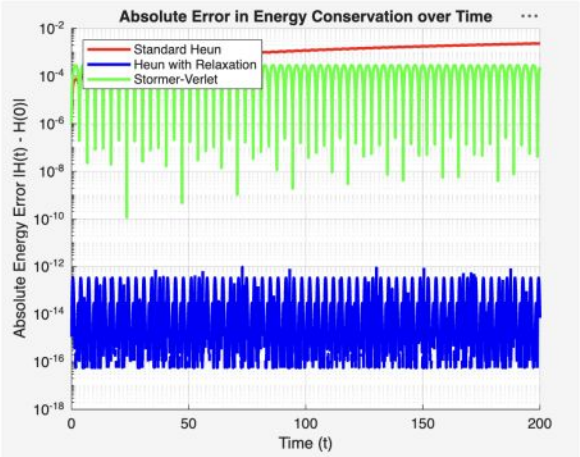
(a) Absolute Area Error (Linear, $h = 0.5$)



(b) Absolute Energy Error (Linear, $h = 0.05$)



(c) Absolute Area Error (Log Scale)



(d) Absolute Energy Error (Log Scale)

Figure 5: Comprehensive dual-benchmark comparing Order 2 methods. The left column tracks absolute area error ($t \in [0, 25]$), while the right column tracks absolute energy error ($t \in [0, 200]$). Logarithmic scaling (bottom row) exposes the discrete polygon approximation limits for area conservation and the Newton-Raphson tolerance limits for energy conservation.

11.5 Experiment 5: Comprehensive Comparison of Area and Energy Conservation (Order 4)

Building directly upon the framework established in the previous section, we extend our dual-benchmark evaluation to fourth-order numerical integrators. We compare the long-term conservation properties of the non-symplectic Standard RK4 method, the symplectic 2-stage Gauss-Legendre method (Gauss 4), and the energy-preserving RK4 with Relaxation. Like the previous experiment, this analysis tracks both absolute area error and absolute energy error on linear and logarithmic scales to expose the underlying structural dynamics.

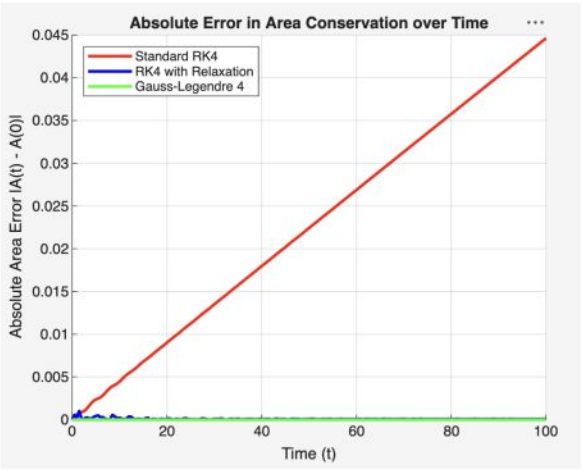
For the geometric area benchmark, we retain the identical initial rectangular domain in phase space ($q \in [-1, 1], p \in [0.5, 1.5]$ with an initial area of 2.0000). To maintain high precision over the long simulation, the boundary is discretized into 50,000 points per edge, yielding a 200,000-sided polygon. The system is integrated from $t = 0$ to $t = 100$ using a step size of $h = 0.5$.

For the physical energy benchmark, we trace the single state trajectory initialized at $q_0 = 0.0$ and $p_0 = 1.0$. This trajectory is integrated over the time domain from $t = 0$ to $t = 100$ using a step size of

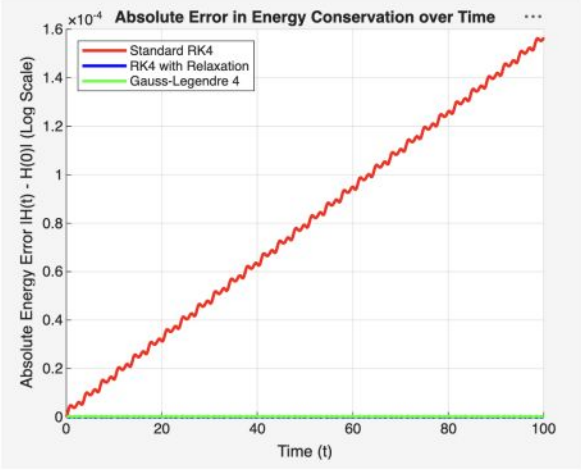
$h = 0.2$.

The experimental results for the fourth-order methods mirror the theoretical dichotomy observed in the second-order methods, albeit at higher baseline precision:

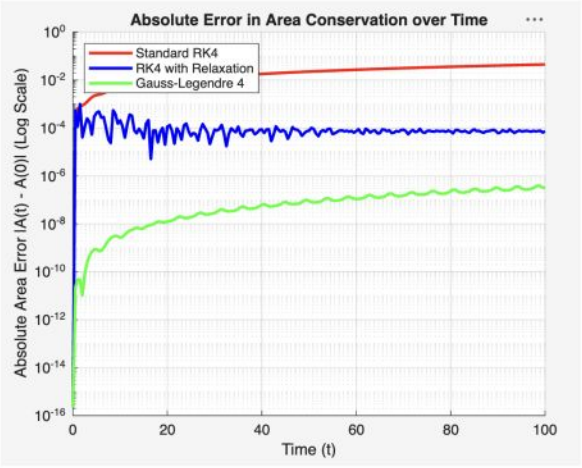
- **Area Conservation Dynamics:** The non-symplectic Standard RK4 method exhibited increasing area error over the tested interval, reaching nearly 0.045 by $t = 100$. The energy-preserving RK4 with Relaxation displays an initial transient deformation but oscillates in a bounded envelope near 10^{-4} on the logarithmic scale, arresting secular phase-space drift over the simulated interval. The symplectic Gauss-Legendre 4 method preserves area in exact arithmetic; its linear plot registers as flat at zero. However, the logarithmic scale exposes a residual error drifting towards the order of 10^{-6}. As noted previously, this is a geometric artifact of the discrete `polyarea` function approximating a continuously warping manifold with a finite polygon, rather than a failure of the symplectic integrator.
- **Energy Conservation Dynamics:** The Standard RK4 method exhibited increasing secular energy drift over the tested interval, with its error accumulating to approximately 1.6×10^{-4}. The symplectic Gauss-Legendre 4 method conserves a nearby shadow Hamiltonian, bounding its energy error oscillations between 10^{-10} and 10^{-6} on the logarithmic scale and showed no visible secular drift. Finally, the RK4 with Relaxation scheme enforces energy preservation at each step; on the logarithmic scale, its energy error drops near machine precision, fluctuating between 10^{-16} and 10^{-12}. These fluctuations represent the numerical noise generated by the finite tolerance of the Newton-Raphson solver when calculating the nonlinear parameter γ.



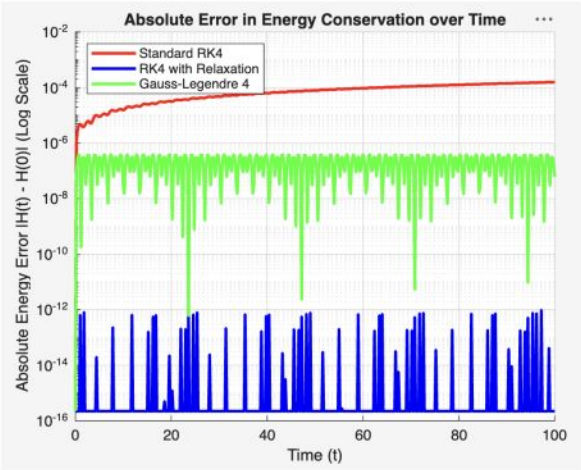
(a) Absolute Area Error (Linear, $h = 0.5$)



(b) Absolute Energy Error (Linear, $h = 0.2$)



(c) Absolute Area Error (Log Scale)



(d) Absolute Energy Error (Log Scale)

Figure 6: Comprehensive dual-benchmark comparing Order 4 methods. The left column tracks absolute area error ($t \in [0, 100]$), while the right column tracks absolute energy error ($t \in [0, 100]$). Logarithmic scaling isolates the discrete polygon approximation limits and the Newton-Raphson tolerance limits at extreme precisions.

12 Conclusion

This project examined structure-preserving numerical methods for Hamiltonian systems through theoretical analysis and numerical experiments on the mathematical pendulum. The study compared standard Runge–Kutta methods, symplectic integrators, and relaxation-based methods using energy conservation and phase-space area preservation as measures of their performance.

The numerical results showed that relaxation methods conserve the Hamiltonian to the precision of the nonlinear solver and can improve phase-space area conservation. Symplectic methods preserve phase-space area theoretically and can improve energy conservation over the simulated interval. These findings show that the benefits of relaxation and symplecticity can extend beyond their primary conservation properties.

An important direction for future work is to investigate whether combining relaxation with symplecticity can improve the preservation of both energy and phase-space area. Further experiments with different Hamiltonian systems, step sizes, and simulation durations are needed to assess the effectiveness and limitations of such methods.

References

[1] E. Hairer, C. Lubich, and G. Wanner, *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*, 2nd ed., Springer Series in Computational Mathematics, Vol. 31, Springer-Verlag, Berlin Heidelberg, 2006.

[2] D. I. Ketcheson, *Relaxation Runge–Kutta Methods: Conservation and Stability for Inner-Product Norms*, SIAM Journal on Numerical Analysis, 57(6):2570–2590, 2019.

[3] *Structure-Preserving Numerical Methods for Ordinary Differential Equations — Code Repository*, GitHub, 2026. Available at: `https://github.com/Tocoung/UGP`.